Theory: where the input should enter a learned KKL observer, why a short input window suffices, and what learned observers can guarantee

draft, generated from theory/m1–m3

Reading guide. Every statement carries a status tag: [proven] under the stated assumptions, [proven, linear plants], [conjecture], or [empirical]; each is followed by a one-sentence intuition, a proof, what it predicts and which experiment node tests it (or that none does). In the paper the theory lives in the appendix; three statements go to the main text: Theorem 12 (why a short window suffices; two-sided for linear plants, with the practical rule of Corollary 13), Proposition 7 (where the input should enter: a stability margin that depends on the input against extrapolation at unseen contexts), and Proposition 23 (when a learned observer beats a filter: a data-independent bias–variance floor under model incompleteness against the learned fit error on the true plant). Proposition 19 is a decomposition with measured constants, not a certificate, and is presented as such. All numerical constants quoted are produced by `theory/checks/*.py`.

1 Where the input enters a learned KKL observer

Setting and conventions. Plant $\dot{x} = f(x,u)$, $y = h(x)$, $x \in \mathbb{R}^{n_x}$, $y \in \mathbb{R}^{n_y}$; no input-affine structure. Inputs are applied through a zero-order hold at the sample rate with values in a compact $U \ni 0$, $\bar{u} := \max_U |u|$; u_t is the window $u|_{[t-L,t]}$ shifted to $[-L, 0]$, a point of $W_L = U^{N_L}$ at sample instants when $L/dt = N_L \in \mathbb{N}$ (the decoder runs only there), with the sup norm. $X(x,t;s;u)$ is the value at s of the solution through x at t. The latent filter is $\dot{z} = Az + By$. *Latent initial condition (one convention for Sections 1–3).* The *label* and training latents start from $z(0) = 0$, so $z(t) = T^0(x(t),t;u) := \int_0^t e^{A(t-s)} B h(X(x(t),t;s;u))\, ds$ exactly; this finite-start map is the only time-varying map used. The *deployed* observer (decoder-side and latent-injection alike, `evaluate.py z0_map`) starts from $\hat{z}(0) = \hat{T}_0(\hat{x}_0)$, which adds the transient $e^{At}\hat{T}_0(\hat{x}_0)$ to the latent; it is carried as the Δz_0 term of Proposition 19 and never absorbed into a map. The decoder target is the windowed map

$$T_L(x,w) := \int_{-L}^{0} e^{-A\tau} B\, h\big(X(x,0;\tau;w)\big)\, d\tau, \tag{1}$$

which by Lemma 11(a) is within $\frac{1}{2}\varepsilon(L)$ of $T^0(x(t),t;u)$ at $w = u_t$. $\mathcal{P}_L$ is the set of pairs $(x(t), u_t)$, $t \geq L$, generated by admissible trajectories.

Assumption 1 (standing; shared by all sections)**.** Regularity: $f(\cdot,u) \in C^1$ uniformly in $u \in U$, $h \in C^2$.

(A1) *Forward invariance.* A compact $\mathcal{X}$ contains every solution from the initial set $\mathcal{X}_0 \subset \mathcal{X}$ under every admissible input, for all $t \geq 0$.

(A2) *Bounded output.* $\|h(x)\| \le \bar{y}$ on $\mathcal{X}$.

(A3) *Hurwitz filter.* $-\alpha < 0$ is the spectral abscissa of A and $\kappa := \sup_{t\ge 0} e^{\alpha t}\|e^{At}\| < \infty$ ($\kappa = 1$ for the normal Butterworth A of the code).

(A4) *Uniform Lipschitz injectivity of the windowed map.* There are L_0 and $\ell < \infty$ with $|x_1 - x_2| \le \ell\,|T_L(x_1, w) - T_L(x_2, w)|$ for all $(x_1, w), (x_2, w) \in \overline{\mathcal{P}_L}$ and all $L \ge L_0$ (i.e. $\ell := \sup_{L\ge L_0} \ell(L) < \infty$; Theorem 12(a) varies L at fixed ℓ).

Chaotic plants. (A1)–(A3) hold for the dissipative chaotic benchmarks (Lorenz, Rössler), (A4) is assumed for them as for every other plant, but its constant is not under control when α is below the *backward* expansion rate λ_b of the flow (for Lorenz $\lambda_b \approx 14.6$ against $\alpha = 2$): $\partial_x T^0$ contains $\int_0^t e^{A(t-s)} B\,\partial h\,\partial_x X(x, t; s; u)\,ds$ and $\|\partial_x X\|$ grows like $e^{\lambda_b(t-s)}$ along the backward flow, so the forward Lipschitz constant of $T^0(\cdot, t)$ grows with t when $\alpha < \lambda_b$; this is the eigenvalue condition of Andrieu and Praly [2] (the spectrum of A must dominate the Lipschitz constant of f for a C^1 stationary map; a Hölder map needs less), and it is a plausible mechanism (not tested: no node sweeps α or measures λ_b) behind the phase-1 PDE residual $\varepsilon_{\text{pde,rms}} = 74$ on Lorenz against ≤ 0.3 elsewhere. Which results survive on such plants is stated at each result: everything that uses only the true past and (A1)–(A2) does (Lemma 11(a), Theorem 12(a), Proposition 19); the counterfactual-past statements (Lemma 11(b)) and the linear lower bounds (Theorem 12(b), Proposition 14(b)) do not apply to them.

Remark 1 (existence and injectivity: cited, and what is assumed)**.** T^0 is a C^1 solution of $\partial_x T\, f(x, u(t)) + \partial_t T = AT + Bh(x)$ whenever backward solutions exist on $[0, t]$ [3, Lemma 1], and an observer $\hat{x} = T^*(\hat{z}, t)$ converges if there are $\bar{t}$ and a concave class-$\mathcal{K}_\infty$ ρ_T with $|x_1 - x_2| \le \rho_T(|T(x_1, t) - T(x_2, t)|)$ for all $t \ge \bar{t}$ [3, Theorem 1]. For a fixed input that is backward distinguishable in finite time, $T^0(\cdot, t)$ is injective for almost every complex diagonal A with $n_y(n_x + 1)$ eigenvalues [3, Theorem 3], but the exceptional set depends on the input [3, Remark 5]; the Lipschitz, uniform-in-time version [3, Theorem 2] needs strong differential observability, inputs of class C^{m+1} (their time derivatives enter) and a gain k above a threshold, which excludes held, switching and PRBS inputs and a fixed (A, B). For autonomous systems, $2n_x + 1$ generic real dimensions suffice [5, Theorem 3.4], $2n_x + 2$ for time-varying ones [5, Remark 3.6]. (A4) is therefore *assumed*: it is the quantitative, uniform-in-input version of what these results give for each regular input. The code's $n_z = 5$ for 2-D plants is the autonomous count; the time-varying count $2n_x + 2$ is not met, so (A4) at $n_z = 5$ is an assumption whose constant must be reported, not a consequence of the cited results; the phase-1 proxy for ℓ (`l_dec_q95`, Remark 25) is 24 (vdp), 44 (bioreactor), 299 (duffing), 10 (lorenz), 14 (quad_tank), 816 (battery) for the C1-where phase 1 (used in `check_apriori_vacuity`); the C2-window-a1 phase 1 of the results layer gives $23, 44, 295, 10, 14, 806$ (Table 1).

Remark 2 (Koopman reading; which model classes the variants are)**.** For the autonomous plant and a diagonalised filter $A = \operatorname{diag}(\lambda_i)$, $B = (b_i)$, the KKL PDE $\partial_x T_i\, f = \lambda_i T_i + b_i h$ reads $(L_f - \lambda_i)T_i = b_i h$ with $L_f = f \cdot \nabla$ the Koopman generator on observables, so

$$T_i = b_i\,(L_f - \lambda_i)^{-1} h = b_i \int_0^\infty e^{\lambda_i \tau}\, h\big(X(x, 0; -\tau; 0)\big)\, d\tau,$$

the resolvent of the generator at the filter pole, i.e. the Laplace transform of the backward Koopman semigroup applied to the output observable; if $h = \sum_j c_j \varphi_j$ with $L_f \varphi_j = \mu_j \varphi_j$ then $T_i = \sum_j b_i c_j \varphi_j / (\mu_j - \lambda_i)$. For a linear plant this is the row identity $p_i = b_i C (F - \lambda_i I)^{-1}$ of the Sylvester solution (check [7]); for a nonlinear plant it gives the spectral reading of the "A fast enough" condition above ($\operatorname{Re}\lambda_i$ must clear the backward spectrum) and of the zero-input floor ε_0 of Remark 6(i)

as a spectral truncation. Under an input the same object is the resolvent for the skew-product (state $\times$ input-shift) system, and the windowed map T_L is its finite-horizon version with the window as a truncated coordinate on the shift space; for a linear plant the context $\dot{q} = Aq - PGu$ of Proposition 5(a) has the structure of a linear predictor with inputs [14, 9]. In these terms, *latent injection* with a stationary T_0 is a lifted Koopman model with inputs (its input-affine case $\Omega(z)u$, and the NLOX baseline $\omega(z)u$, are bilinear lifted observers with a fixed output-injection gain B), whose stability requires the input-dependent generator $A + \partial_z \Phi$ to stay contracting (Proposition 7(a)); *decoder-side conditioning* keeps a fixed generator and parametrises the left inverse of the lifting by the input context. Proposition 9 is then the statement that the Koopman spectrum moves with u (eigenvalue $u - a$; rotation rate $\omega_0 + u$) while a fixed A does not. We state this as positioning only; no result below depends on it.

1.1 The decoder target and sufficient contexts

Definition 3 (sufficient context)**.** A context map $E : W_L \to \mathbb{R}^{d_c}$ is *sufficient* if the latent and the context determine the state: for $(x_i, w_i) \in \overline{\mathcal{P}_L}$, $E(w_1) = E(w_2) \Rightarrow |x_1 - x_2| \le \ell\, |T_L(x_1, w_1) - T_L(x_2, w_2)|$. It is ϵ_E-sufficient if the same holds with an additive ϵ_E on the right.

The window itself ($E = \mathrm{id}$) is sufficient by (A4). A lower-dimensional E cannot be injective on W_L (invariance of domain), but it need not be: it only has to resolve what the latent leaves open.

Lemma 4 (decoder target)**.** *Let E be continuous and ϵ_E-sufficient, and $S_L := \{(T_L(x, w), E(w)) : (x, w) \in \overline{\mathcal{P}_L}\}$. If $\epsilon_E = 0$ there is a continuous $G^* : S_L \to \mathcal{X}$ with $G^*(T_L(x, w), E(w)) = x$, ℓ-Lipschitz in z for each c; extend $G^*(\cdot, c)$ to $\mathbb{R}^{n_z}$ with the same constant (Kirszbraun). Then along every admissible trajectory, for $t \ge L$, $|G^*(z(t), E(u_t)) - x(t)| \le \frac{\ell}{2}\varepsilon(L)$. For $\epsilon_E > 0$, any decoder D satisfies $\sup |D(T_L(x, w), E(w)) - x| \ge \frac{1}{2}\delta_E$, where δ_E is the largest $|x_1 - x_2|$ over pairs with equal latent and equal context.*

Intuition. Over a fixed window the plant has no clock, so the state is a fixed function of (latent, context) as soon as the context resolves the latent's ambiguity.

Proof. $\Psi(x, w) = (T_L(x, w), E(w))$ is continuous on the compact $\overline{\mathcal{P}_L}$ (under the hold, the backward solution is a composition of N_L C^1 flows with constant parameters). By sufficiency, $\Psi(x_1, w_1) = \Psi(x_2, w_2)$ implies $x_1 = x_2$, so π_x factors as $G^* \circ \Psi$; G^* is continuous because Ψ is a closed map (continuous on a compact set into a Hausdorff space), and ℓ-Lipschitz in z by sufficiency. Then $|G^*(z(t), c) - x(t)| = |G^*(z(t), c) - G^*(T_L(x(t), u_t), c)| \le \ell\, |z(t) - T_L(x(t), u_t)|$ and Lemma 11(a). The lower bound: D returns one value for two states δ_E apart. □

Proposition 5 (sufficient contexts exist, and must carry history)**.** *(a)* Linear plants. *For $\dot{x} = Fx + Gu$, $y = Cx$, the windowed map is affine in x with a w-independent slope, $T_L(x, w) = P_L x + Q_L w$, $P_L = P - e^{AL} P e^{-FL}$, $PF = AP + BC$. Hence the n_z-dimensional linear context $c = Q_L w$ is exactly sufficient whenever P_L is injective ($\ell = \|P_L^+\|$), and as $L \to \infty$ (under the washout $e^{AL} P e^{-FL} \to 0$, (H1) of Theorem 12) it is the output q of the stable filter $\dot{q} = Aq - PGu$: a spectral filter bank with identity readout. (b)* Nonlinear plants. *The window ($d_c = n_u N_L$) is sufficient, and by (a) of Lemma 11 any context that determines u_t for some L is $\ell\varepsilon(L)$-sufficient for the infinite past: $|x_1 - x_2| \le \ell\, |T_1^0 - T_2^0| + \ell\, \varepsilon(L)$ (each T_i^0 is within $\frac{1}{2}\varepsilon(L)$ of T_L). (c)* History is necessary. *The current input alone is not sufficient in general (Proposition 14).*

Proof. (a) $X(x, 0; \tau; w) = e^{F\tau} x +$ (linear in w); substitute in (1) and use $\int_{-L}^{0} e^{-A\tau} B C e^{F\tau} d\tau = P - e^{AL} P e^{-FL}$ (integrate $\frac{d}{d\tau}(e^{-A\tau} P e^{F\tau}) = e^{-A\tau} B C e^{F\tau}$). If $Q_L w_1 = Q_L w_2$ then $T_L(x_1, w_1) -$

$T_L(x_2, w_2) = P_L(x_1 - x_2)$. The limit: $z - Px$ obeys $\dot{(z - Px)} = A(z - Px) - PGu$. (b) (A4) and Lemma 11(a). □

For nonlinear plants and contexts of fixed dimension below $n_u N_L$, finite stable filter banks with a static readout uniformly approximate every time-invariant fading-memory operator on slew-limited bounded inputs [4, Theorem 2]; applying it to the observer operator $(y, u) \mapsto x$ (a fading-memory operator under (A1)–(A4), on slew-limited signals, and for the filters the theorem chooses rather than ours) motivates the spectral context but is not a guarantee for the trained encoder.

Remark 6 (expressivity at a fixed base; what C1 compares)**.** (i) *Shared floor.* Every conditioner with exact zero-input collapse, $D(\cdot, 0) = T_0^+$, errs at $c = 0$ by $\varepsilon_0 := \sup_z |G^*(z, 0) - T_0^+(z)|$: the zero-input target is the inverse of the zero-input windowed map, not the learned stationary base. (ii) *Additive (concat).* $T_0^+(z) + R(z, c) - R(z, 0)$ approximates on S_L, as the width of R grows, every continuous target that equals T_0^+ on the zero slice: SiLU satisfies $\mathrm{SiLU}(s) - \mathrm{SiLU}(-s) = s$, so a two-hidden-layer SiLU network contains every one-hidden-layer one, and the universal approximation theorem for non-polynomial activations [10] applies (with a Tietze extension of the target). (iii) *Base-capped (FiLM, LoRA, hypernetwork).* For each fixed c the decoder is a network $N_{\theta(c)}$ of the frozen base's architecture, so for every conditioner budget

$$\inf_{D} \sup_{S_L} |D - G^*| \;\geq\; \sup_{c} \inf_{\theta \in \Theta} \sup_{z} |N_\theta(z) - G^*(z, c)|,$$

with Θ the reachable parameter set. At the matched budget Θ is much smaller than the base's weight space: the hypernetwork's linear bias-free heads place $\theta(c)$ in an affine subspace of dimension equal to its trunk width; LoRA updates have the configured rank and leave biases fixed; FiLM never modulates the readout layer, so $D(\cdot, c) = W_L \phi_c(z) + b_L$ with (W_L, b_L) frozen. In the ReLU setting, a LoRA rank of order width times a depth ratio suffices to represent a smaller target network exactly [18]; that result does not cover tanh bases or small fixed ranks. (iv) *Latent injection.* Φ is a universal class, but as an observer it represents only targets of the stationary route: a stationary T_0 with an exact correction. The theory therefore does not rank the variants by expressivity; at a matched budget on one base, C1 measures inductive bias and optimisation within these classes.

1.2 Where the input enters: a stability–extrapolation trade-off

Both observers run the filter on $y + \nu$, $\|\nu\|_\infty \leq \delta_y$, with a context $\hat{c} = E(\hat{u})$ of the input $\hat{u}$ they are given; $c = E(u)$ and $\delta_c := \sup_t |\hat{c} - c| \leq \ell_E \|\hat{u} - u\|_\infty$ (Lemma 15).

$$\begin{aligned} &\text{(latent)} \quad \dot{\hat{z}} = A\hat{z} + B(y + \nu) + \Phi(\hat{z}, \hat{c}), && \hat{x} = T_0^+(\hat{z}), \quad \hat{z}(0) = T_0(\hat{x}_0), \\ &\text{(decoder side)} \quad \dot{\hat{z}} = A\hat{z} + B(y + \nu), && \hat{x} = D(\hat{z}, \hat{c}), \quad \hat{z}(0) = \hat{T}_0(\hat{x}_0). \end{aligned}$$

For the latent variant, $T_0 \in C^1$ is a stationary map with left inverse T_0^+ (ℓ_{dec}-Lipschitz, $\varepsilon_{\mathrm{dec}} = \sup_{\mathcal{X}} |T_0^+ T_0(x) - x|$), the residual along trajectories is $r = \partial_x T_0(x) f(x, u) - A T_0(x) - B h(x) - \Phi(T_0(x), c)$ with $|r| \leq \varepsilon_{\mathrm{res}}$, and $\ell_{\Phi,c}$ is the Lipschitz constant of $\Phi(z, \cdot)$ *uniformly over* $z \in T_0(\mathcal{X}) + B(r)$ (the set used in (a)). $\mu_2(M) = \lambda_{\max}(\frac{1}{2}(M + M^\top))$ is the logarithmic norm.

Proposition 7 (where the input enters)**.** *Under Assumption 1:*

(a) Latent injection: stability depends on the input *[proven]. Let* $\bar{d} := \varepsilon_{\mathrm{res}} + \|B\|\delta_y + \ell_{\Phi,c}\,\delta_c$ *and let* $r \geq |e(0)| + \bar{d}/\mu$. *If* $\mu_2\big(A + \partial_z \Phi(z, \hat{c})\big) \leq -\mu < 0$ *for all* z *in the set* $\mathcal{Z}_r := T_0(\mathcal{X}) + B(r)$ *and all contexts* $\hat{c}$ *the observer meets, then* $e = \hat{z} - T_0(x)$ *stays in* $B(r)$ *for all* t *(so the hypothesis is checked on a set fixed in advance, not on the solution) and*

$$|e(t)| \leq e^{-\mu t}|e(0)| + \tfrac{1}{\mu}\big(\varepsilon_{\mathrm{res}} + \|B\|\delta_y + \ell_{\Phi,c}\,\delta_c\big), \qquad |\hat{x} - x| \leq \ell_{\mathrm{dec}}|e| + \varepsilon_{\mathrm{dec}}.$$

(b) Decoder side: stability is free, extrapolation is not *[proven]. The latent is an input-free LTI filter, so $\hat{z} - z$ is driven only by ν and the initial transient, and $|\hat{x} - x|$ obeys the bound of Proposition 19 for every input, with the decoder's Jacobian norms $\ell_z(\hat{c}), \ell_c(\hat{c})$ taken at the contexts actually queried. Both conditioner families have c-uniform bounds on a bounded context set, but with very different constants: for an additive conditioner $\ell_z(c) \le \ell_z(T_0^+) + 2\sup\|\partial_z R\|$; for multiplicative ones (FiLM, LoRA, hypernetwork) the available bound is the product form $\ell_z(c) \le \prod_l \|W_l + \Delta W_l(c)\| \sup|\sigma'|^d$, finite because $\theta(c)$ is bounded (tanh trunk, bounded c), but 10^1–10^5 times larger at random initialisation (check [5]) before the tanh base saturates.*

Intuition. Injecting the input makes the observer a recurrent network whose stability depends on how strongly the injection reacts to the latent, but its decoder never sees an unfamiliar context; conditioning the decoder keeps the recurrent part a fixed stable filter, but the network is queried at every unseen context directly.

Proof. (a) $\dot{e} = Ae + [\Phi(\hat{z}, \hat{c}) - \Phi(T_0 x, \hat{c})] + d$, $d = B\nu + [\Phi(T_0 x, \hat{c}) - \Phi(T_0 x, c)] - r$, $|d| \le \bar{d}$. While $e \in B(r)$ the segment $[T_0 x, \hat{z}]$ lies in the ball $T_0 x + B(r) \subseteq \mathcal{Z}_r$ (convexity of the ball, not of $\mathcal{Z}_r$, is what is used); write the bracket as $\bar{J}e$, $\bar{J} = \int_0^1 \partial_z \Phi(T_0 x + se, \hat{c}) ds$, so $\mu_2(A + \bar{J}) \le -\mu$ by subadditivity and convexity of μ_2, and $D^+|e| \le -\mu|e| + \bar{d}$. The comparison function $\phi(t) = e^{-\mu t}|e(0)| + \bar{d}(1 - e^{-\mu t})/\mu \le r$ bounds $|e|$ as long as $e \in B(r)$, and $\phi < r$ whenever $|e| = r$ would be reached, so $B(r)$ is forward invariant and the bound holds for all t (contraction, 11; equivalently the perturbation lemmas of 8, Ch. 9). The state bound is the triangle inequality through $T_0^+(T_0(x))$. (b) The latent statement is linearity; the bound is Proposition 19; the Jacobian bounds are the chain rule (R's z-Jacobian does not involve c multiplicatively, whereas modulated layers multiply z-features by c-dependent weights), and $\theta(c)$ is bounded because the trunk is tanh and c is bounded (Lemma 15). □

Corollary 8 (the certified margin shrinks with the context)**.** *$\mu_2(A + \partial_z \Phi) \le \mu_2(A) + \ell_\Phi$, and $\mu_2(A) = -\alpha$ for normal A, so $\mu \ge \alpha - \ell_\Phi$ (for general A, the Gronwall version gives $\alpha - \kappa\ell_\Phi$). For an injection with exact zero-input collapse, $\Phi(z, c) = M(z, c) - M(z, 0)$, $M \in C^2$: $\ell_\Phi(c) \le \min\big(|c|\, L_{zc}, 2\sup\|\partial_z M\|\big)$ with $L_{zc} = \sup\|\partial_c \partial_z M\|$, and $|c| \le \ell_E \|u\|_\infty$ (sup norm) since $E(0) = 0$; for an input-affine injection $\Omega(z)u$, $\ell_\Phi \le \ell_\Omega \|u\|_\infty$. The margin is evaluated at the* observer's *context $\hat{c}$, so a corrupted input shrinks it. The margin saturates: if $2\kappa \sup\|\partial_z M\| < \alpha$ the latent observer is certified at every amplitude, and when the plant input saturates (as through a saturating actuator) the exact correction's z-Lipschitz constant stays bounded too. The condition is* sufficient only*: for the code's trained injection the certified margin is already negative at moderate gain (-5.47, log-norm $+8.1$ in check [3]) while two observer copies still contract, so a negative margin predicts nothing, and only a measured incremental rate does.*

Proposition 9 (the margin cannot be removed at a fixed spectrum)**.** *[proven] There are bounded, observable plants with an* exact *injection ($\varepsilon_{\text{res}} = 0$) whose latent observer diverges while a decoder-side observer with a bounded sufficient context is within $\frac{\ell}{2}\varepsilon(L)$ of exact. For $\dot{x} = (u - a)x + b$, $y = x$, $a > \alpha > 0$, scalar latent $\dot{z} = -\alpha z + y$ and constant $u \in (\alpha, a)$: the plant converges to $b/(a - u)$, the stationary map is $T_0(x) = Px + q$, $P = 1/(\alpha - a)$, $q = -Pb/\alpha$, the exact injection is $\Phi^*(z, u) = u(z - q)$ ($\ell_\Phi = |u|$), and $\dot{e} = (u - \alpha)e$. The windowed map is $T_L(x, w) = P_L(u)\, x + Q_L(u)$ with $P_L = \int_{-L}^0 e^{(\alpha + u - a)\tau} d\tau > 0$ and $Q_L = \int_{-L}^0 e^{\alpha\tau}\, b\, \frac{e^{(u-a)\tau} - 1}{u - a}\, d\tau$ (both bounded functions of the window input), so the decoder $\hat{x} = (\hat{z} - Q_L)/P_L$ errs by at most $\frac{\ell}{2}\varepsilon(L)$ with $\ell = 1/P_L$ (Lemma 4), for every $u \in (\alpha, a)$.*

Proof. $T_0'(x)f(x,0) = P(b-ax)$ and $-\alpha T_0 + x = (1-\alpha P)x - \alpha q = -aPx + Pb$; so T_0 solves the zero-input PDE and $\partial_x T_0[f(x,u) - f(x,0)] = Pux = u(T_0(x) - q)$. Then $\dot{e} = -\alpha\hat{z} + x + u(\hat{z} - q) - P((u-a)x + b) = (u-\alpha)e$. For the window: $X(x,0;\tau;u) = e^{(u-a)\tau}x + b(e^{(u-a)\tau} - 1)/(u-a)$ in (1). □

Here the small-gain threshold $u = \alpha$ is attained exactly. The *exact* time-varying map $z = m(t)x + n(t)$, $\dot{m} = 1 - (\alpha + u - a)m$, $\dot{n} = -\alpha n - bm$, is also a sufficient context, but for $u < a - \alpha$ its coefficient $-(\alpha + u - a)$ is positive and $m, n \to \infty$ (check: $m(30) = 2 \cdot 10^{19}$ at $u = 0.5$); it is outside the bounded class of Proposition 7(b), which is why the windowed context is the one used. The obstruction is intrinsic to the stationary route *at a fixed spectrum*: a designer removes it by choosing $\alpha > \bar{u}$ (here) or $\alpha > \bar{u}\ell_\Omega\kappa$ (Corollary 8), i.e. by a gain that grows with the input bound, which is also what Bernard and Andrieu [3, Theorem 4] require. A second example, the rotating oscillator $\dot{x} = (\omega_0 + u)Jx$, $y = x_1$ ($|x|$ is conserved for every input), with the drift map P and the exact correction $\Omega(z) = PJP^+z$, has error rate $\max \mathrm{Re}\,\lambda(A + uPJP^+)$: stable for all tested $u > -\omega_0$, but unstable ($+2.65$, $+5.16$ at $u = -2, -4$, $\omega_0 = 1$, $n_z = 5$, $\alpha = 1.5$) when the rotation reverses, although the plant is bounded and observable there; this is the input-independence required by the stationary route [3, Theorem 4], which also needs an input-affine plant and a gain growing with $\bar{u}$. The decoder-side route needs only Assumption 1 and a sufficient context.

What the bounds predict, and what the current nodes test. Under protocol v2 no tier changes the input amplitude (every family lives in the same box), so the quantity that varies between tiers is the *distance in context space*, $d(\hat{c}, \mathcal{C}_{\text{train}})$, not $|c|$. (P1) The latent variant's excess error on `shape_ood` and `sequence_ood` is the extrapolation error of Φ at the new contexts, filtered at the measured incremental rate; it degrades sharply only if that rate approaches zero. Decoder-side variants pay $\ell_z(\hat{c})\,|\Delta z|$ and $\ell_c(\hat{c})\,|\Delta c|$ at the new contexts (Corollary 10), with the additive conditioner's ℓ_z small and c-uniform. (P2) Input noise moves the latent variant's operating point to a corrupted context $\hat{c}$ (its margin is evaluated there), whereas decoder-side variants stay stable for every input and pay $\ell_c\ell_E\delta_u$. *Status: untested.* No current node measures μ_2 or the incremental rate on visited segments, $\ell_z(\hat{c})$, $\ell_c(\hat{c})$, or the context distance per tier; under C3-faults `input_noise` every variant, latent included, degrades by about the same factor ($\approx 10\times$ on vdp, read from the C3-faults node report; not a number of the generated results layer), which neither supports nor falsifies (P2). The quantities to record are eval-only on saved checkpoints: $\min_{c' \in \mathcal{C}_{\text{train}}} \|\hat{c} - c'\|$ per tier (C1-where, TEST-C1), the latent margin $\alpha - \kappa\ell_\Phi$ already computed by `metrics.py`, and the realised gains of Definition 24. Measured Lipschitz constants and log-norms are estimates on the visited set (lower bounds of suprema), not certificates.

Corollary 10 (out of distribution: Lipschitz extrapolation, and nothing more)**.** *[proven] Let* $\varepsilon_{\text{train}} := \sup |D(z', E(w')) - x'|$ *over the training support* S_{train} *of triples (label latent* $z' = T^0$*, window* w'*, state* x'*), and* assume *the target* $x = G^*(z,w)$ *of Lemma 4 has, on its domain* S_L*, Lipschitz constants* ℓ *in* z *(this is (A4)) and* ℓ_w *in* w *(sup norm; a new assumption, not implied by sufficiency), and assume its extension beyond* S_L *keeps these constants (a componentwise McShane extension achieves them only up to a factor* $\sqrt{n_x}$*, which then multiplies* ℓ *and* ℓ_w *in (a)). (a) For a query* (z,w) *and any* $(z',w') \in S_{\text{train}}$*, with* D*'s Jacobian norms* ℓ_z^D, ℓ_c^D *on the segment,*

$$|D(z, E(w)) - G^*(z,w)| \le \varepsilon_{\text{train}} + \tfrac{\ell}{2}\varepsilon(L) + (\ell_z^D + \ell)|z - z'| + (\ell_c^D\,\ell_E + \ell_w)\|w - w'\|_\infty,$$

the $\frac{\ell}{2}\varepsilon(L)$ *because* $G^*(z',w') = x'$ *only at* $z' = T_L$*, whereas the training latent is* T^0*; and* $|G^*(z(t), u_t) - x(t)| \le \frac{\ell}{2}\varepsilon(L)$ *(Lemma 4). (b) The linear rate cannot be improved without structure. Among targets that are* 2λ*-Lipschitz and agree with a* λ*-Lipschitz one on* S_{train}*,* $G_\pm = G_0 \pm \lambda\, d(\cdot, S_{\text{train}})\, e$ *(*$|e| = 1$*)*

both fit the data, so every learner errs by at least $\lambda\, d(q, S_{\text{train}})$ at some query q. At the plant level, if φ is a diffeomorphism of $\mathcal{X}$ with $h \circ \varphi = h$, equal to the identity on an open neighbourhood of the states reached under the training inputs (so that $D\varphi = I$ there), then the plants f and $\varphi_* f$ produce identical training data, and under an input that drives the state where $\varphi \neq \mathrm{id}$ they produce identical (y, u) from x_0 and $\varphi(x_0)$, so any observer errs by at least $\frac{1}{2}|\varphi(x(t)) - x(t)|$ on one of them.

Proof. (a) Triangle inequality through $D(z', E(w'))$, $G^*(z', w')$ and $G^*(T_L(x', w'), w') = x'$, with $|E(w) - E(w')| \le \ell_E \|w - w'\|_\infty$ and $|z' - T_L(x', w')| \le \frac{1}{2}\varepsilon(L)$. (b) $d(\cdot, S)$ is 1-Lipschitz and vanishes on S; $|G_+ - G_-| = 2\lambda d$ and one learner output is at distance $\ge \lambda d$ from one of them. For the plants: φ maps solutions of f to solutions of $\varphi_* f$ and preserves y; on the training set it is the identity. □

What it predicts under protocol v2. Distances are measured in input (window) space so that they are comparable across encoders. In v2 at most 0.1% of evaluation states lie outside the training box on any dev system and tier (C1-where node report; not in the generated results layer), and the input amplitude box is shared by all tiers, so $|z - z'|$ and the state-level construction of (b) are inert: the relevant distance is $\|w - w'\|_\infty$ (equivalently $d(\hat{c}, \mathcal{C}_{\text{train}})$ in context space), which `shape_ood` and `sequence_ood` change through the waveform and the event timing at the same amplitude. The corollary therefore predicts a shape/sequence-OOD excess that is at most linear in that context distance and, by (b), can be no better without structure; it does not predict its size, and the context distance per tier is not yet reported (eval-only; C1-where, TEST-C1). A model-based filter with the correct model extrapolates by structure; a learned observer cannot, beyond this Lipschitz rate.

Zero input. With $E(0) = 0$, $\Delta(\cdot, 0) = 0$ in every decoder conditioner and $\Phi(\cdot, 0) = 0$, every conditioned observer is the autonomous KKL observer at $u \equiv 0$, so the autonomous theory [2, Theorems 2.2, 2.4, 2.7] and the robustness bound of Niazi et al. [13, Theorem 19] (their cond(V) is our κ) are the special case $c \equiv 0$ of Proposition 19.

Numerical checks. `theory/checks/m1_location.py`: [1] zero-input collapse of all encoders and conditioners (errors $\le 5 \cdot 10^{-16}$); [2] at fixed c, FiLM, LoRA and hypernetwork decoders equal base-architecture networks with explicit weights ($\le 3 \cdot 10^{-13}$); [3] the injection's z-Lipschitz constant is $0, 0.61, 1.22, 2.33, 4.26, 6.82, 7.29$ at $|c| = 0, 0.25, 0.5, 1, 2, 8, 32$ (linear, then saturating), and for two observer copies the measured incremental rate is below the sup log-norm on the visited segments ($-1.52 \le -1.23$; $-1.05 \le -0.29$, where the small-gain margin is -5.47 and certifies nothing; at the largest gain the log-norm is $+8.1$ and also certifies nothing while the copies still contract, so the log-norm condition is sufficient only); [4] the rotating oscillator; [5] *at random initialisation* (trained decoders are not tested here) the z-Jacobian of concat stays below its uniform-in-c bound (≤ 1.25 against 12 for $|c| \le 256$) while FiLM, LoRA and hypernetwork reach 81, $3.7 \cdot 10^5$, $2.3 \cdot 10^6$ and then fall (LoRA and hyper to 0 and 2.9 at $|c| = 256$): c-uniformly bounded, with a far larger constant, as stated in Proposition 7(b); [6] the linear windowed map is affine with w-independent slope to 10^{-15}; [7] the Koopman resolvent identity $p_i = b_i C (F - \lambda_i I)^{-1}$ (to 10^{-16}) and its Laplace-transform form (to 10^{-8}); [8] the a-priori form of Proposition 7(a): with $\mu = 0.46$, $|e(0)| = 0.5$, $\sup |d| = 0.3$ the error never leaves $B(r)$, $r = 1.15$. `theory/checks/check_theory.py`: `where_input_enters` (Proposition 9: measured rate $u - \alpha$ to 10^{-3}; windowed-context error equals $\frac{\ell}{2}\varepsilon(L)$ to three digits at $L = 6$, i.e. the bound is attained for a constant input; the exact context's $m(t)$ reaches 10^{11}–10^{19}).

2 Why a short input window suffices

Setting. Assumptions (A1)–(A4) of Section 1: forward invariance of a compact $\mathcal{X}$, $\|h\| \le \bar{y}$ on $\mathcal{X}$, $\|e^{At}\| \le \kappa e^{-\alpha t}$, and Lipschitz injectivity (modulus ℓ, uniform in $L \ge L_0$) of the windowed map. Here we also use that $-\alpha$ is *exactly* the spectral abscissa of A and that the eigenvalues on $\operatorname{Re}\lambda = -\alpha$ are semisimple (otherwise no finite κ exists for that α). The Butterworth real-Jordan A is normal: each block is $sI + \omega J$, so $\|e^{At}\| = e^{-\alpha t}$ and $\kappa = 1$; its real parts are $\alpha_j = \alpha \sin\frac{(2j-1)\pi}{2n_z} / \sin\frac{\pi}{2n_z}$, $j = 1, \dots, \lceil n_z/2 \rceil$, so $\alpha = \alpha_1 \le \alpha_j \le \alpha_{\max} = \alpha / \sin\frac{\pi}{2n_z}$. The *label* latent starts at $z(0) = 0$ (the convention of Section 1), hence $z(t) = T^0(x(t), t; u)$ exactly; the deployed observer's initial transient is a separate term (Proposition 19) and plays no role in this section. $T_L(x, w)$ is the windowed map (1) (zero history before the window), and

$$\varepsilon(L) := \frac{2\kappa\|B\|\bar{y}}{\alpha}\, e^{-\alpha L}.$$

Lemma 11 (fading-memory tail)**.** *(a) [proven] Along every admissible trajectory and $t \ge L$, $\|T^0(x(t), t; u) - T_L(x(t), u_t)\| \le \frac{1}{2}\varepsilon(L)$. (b) [proven under its hypothesis] If $u = v$ on $[t-L, t]$ and* both *backward solutions from x stay in $\mathcal{X}$ on $[0, t]$, then*

$$T^0(x,t;u) - T^0(x,t;v) = e^{AL}\big[T^0(\xi, t-L; u) - T^0(\xi, t-L; v)\big], \qquad \|T^0(x,t;u) - T^0(x,t;v)\| \le \varepsilon(L),$$

with ξ the common backward state at $t - L$. In particular the map forgets at exactly rate α: for a unit left eigenvector p of an eigenvalue with real part $-\alpha$, $|p^*[T^0(x,t;u) - T^0(x,t;v)]| = e^{-\alpha L}\,|p^*[T^0(\xi, t-L; u) - T^0(\xi, t-L; v)]|$.

Intuition. Inside the window the two backward solutions are one curve; everything older reaches the latent through e^{AL}.

Proof. (a) $T^0 - T_L$ is the integral over $[0, t-L]$ of $e^{A(t-s)}Bh(x(s))$ along the true past, which lies in $\mathcal{X}$ by (A1); bound the integrand by $\kappa\|B\|\bar{y}e^{-\alpha(t-s)}$. (b) On $[t-L, t]$ both backward solutions solve the same ODE from x, so they coincide; split (1)-type integrals at $t - L$ and use the flow property: the window parts cancel and the rest is $e^{AL}T^0(\xi, t-L; \cdot)$. Bound each map's remainder as in (a). Finally $p^*e^{AL} = e^{\lambda L}p^*$, $|e^{\lambda L}| = e^{-\alpha L}$. □

No Lipschitz property of f in u is used; the plant enters only through $\bar{y}$. *Where (b) does not apply.* Part (a) uses only the true past and (A1), so it holds for every benchmark. Part (b) needs the *counterfactual* backward solution (from x under v) to stay in $\mathcal{X}$; for a dissipative chaotic flow this fails generically, because backward time expands at the rate λ_b of the most contracting forward direction (Lorenz: $\lambda_b \approx 14.6$, so a counterfactual past leaves any compact set within a fraction of a second). For Lorenz and Rössler, (b) and everything that rests on it (Proposition 5(b)'s $\ell\varepsilon(L)$-sufficiency of a context that determines the window; the lower bounds below) are not available; the sufficient direction (a) is. The zero-order hold changes nothing: if the held samples agree for $j = k - M, \dots, k-1$, apply the lemma at t_k with $L = M\,dt$ (the sample u_k has not acted yet); the deployed discrete filter has the same bounds with $e^{-\alpha L} \mapsto e^{-\alpha M dt}$ and constants no larger than the continuous ones, plus e_{disc} (Lemma 26).

Theorem 12 (window rule for the observer)**.** *Let $L^*(\eta)$ be the shortest window for which some decoder $D(z, u_t)$ has worst-case error at most η.*

*(a) (*sufficient; every plant under (A1)–(A4)*) [proven] $D = T_L^*$ achieves $\frac{\ell}{2}\varepsilon(L)$, so*

$$L^*(\eta) \;\le\; \frac{1}{\alpha}\ln\frac{\kappa\ell\|B\|\bar{y}}{\alpha\eta} \;=\; \frac{\ln(1/\eta) + \ln\ell + \ln(\kappa\|B\|\bar{y}/\alpha)}{\alpha}.$$

*(b) (*necessary; proven for linear plants*) For every decoder and every plant,* $\sup \|D-x\| \geq \frac{1}{2}\delta(L)$*, where the fibre diameter* $\delta(L)$ *is the largest* $\|x-x'\|$ *over* admissible *pairs with* $T^0(x,t;u) = T^0(x',t;v)$ *and* $u=v$ *on* $[t-L,t]$ *(each state reached at* t *under its own past). For a linear plant* $\dot{x}=Fx+Gu$*,* $y=Cx$*, normal* A*,* (A,PG) *controllable, symmetric inputs* $|u|\leq\bar{u}$*, under*

(H1) washout*:* $e^{At}Pe^{-Ft}\to 0$*, so that after a transient* $z=Px+q$ *with* $PF=AP+BC$*,* P *injective,* $q(t)=-\int_{-\infty}^{t} e^{A(t-r)}PGu(r)dr$ *(this holds iff* α *exceeds the fastest* contraction *rate of* F*; automatically for neutrally stable* F*);*

(H2) overlapping reachable sets*: for any two admissible input histories and every* d *with* $\|d\|\leq 2\kappa R_q e^{-\alpha L}/\sigma_{\min}(P)$ *there is a state* x *reachable at* t *under the first history with* $x+d$ *reachable under the second. (Every* $x\in\mathcal{X}$ *being reachable under every history is* not *required and is false under (A1) whenever the forced response is nonzero: the reachable set under one history is the translate* $e^{Ft}\mathcal{X}_0+$*(forced response), a strict subset of* $\mathcal{X}$*. (H2) holds for* F *skew-symmetric or* $F=0$ *when* $\mathcal{X}_0$ *contains a ball whose radius exceeds the forced-response bound plus* $\|d\|/2$*, as in the oscillator and integrator examples; it fails for contracting* F*, where the reachable set under a fixed history shrinks to a point and the fibre, hence the necessary window, is* smaller*);*

the fibre diameter satisfies

$$\frac{r_q}{\|P\|}\,e^{-\alpha_k L} \;\leq\; \delta(L) \;\leq\; \frac{2\kappa R_q}{\sigma_{\min}(P)}\,e^{-\alpha L},$$

where $R_q=\kappa\|PG\|\bar{u}/\alpha$*,* $r_q>0$ *is the radius of a ball contained in the set* Q *of reachable* q*, and* α_k *is the real part of the first mode group such that the slowest groups up to it span a dimension* $\geq n_z-n_x+1$*. Hence* $\alpha\leq\alpha_k\leq\alpha_{\max}$ *and*

$$\frac{1}{\alpha_k}\ln\frac{r_q}{2\|P\|\eta} \;\leq\; L^*(\eta) \;\leq\; \frac{1}{\alpha}\ln\frac{\kappa\ell\|B\|\bar{y}}{\alpha\eta}, \qquad L^*(\eta)=\Theta\Big(\frac{1}{\alpha}\ln\frac{1}{\eta}\Big) \quad \textit{(linear plants)},$$

with constants that depend on n_z *only through* $\alpha_k/\alpha\leq 1/\sin\frac{\pi}{2n_z}$ *(*$\alpha_k/\alpha=2.618$ *at* $n_z=5$*,* $n_x=2$*;* 4.049 *at* $n_z=7$*,* $n_x=3$*).*

Intuition. The decoder cannot recover what the window does not contain, and what it does not contain reaches the latent through e^{AL}; a decoder may lean on the faster modes, but never faster than $\alpha_{\max}$.

Proof. (a) Lemma 4 with Lemma 11(a). (b) Two points of the fibre receive the same input (z,u_t), so D errs by $\geq\frac{1}{2}\|x-x'\|$ on one of them. Linear case: by (H1) the window parts of q coincide, so $Px+q=Px'+q'$ iff $P(x-x')=e^{AL}(m'-m)$ with $m,m'\in Q$ the history terms at $t-L$; every such $d=x'-x$ obeys the upper bound below, so by (H2) the pair can be realised by admissible states, and $\delta(L)=\sup\{\|d\| : e^{-AL}Pd\in Q-Q\}$. Upper: $\|d\|\leq\|Pd\|/\sigma_{\min}(P)\leq\|e^{AL}\|\,2R_q/\sigma_{\min}(P)$. Lower: $\dim\operatorname{range}P=n_x$ and the slow span S_k has dimension $\geq n_z-n_x+1$, so some unit d has $Pd\in S_k$; for normal A, $\|e^{-AL}Pd\|\leq e^{\alpha_k L}\|P\|$, and $Q-Q\supseteq Q\supseteq B(r_q)$ ($0\in Q$, controllability), so $d\,r_q e^{-\alpha_k L}/\|P\|$ is feasible. □

What is proven and what is not. The two-sided $\Theta(\ln(1/\eta)/\alpha)$ is a statement about linear plants (with the hidden factor α_k/α in the lower constant at $n_z=5$). For nonlinear plants only (a) is proven; "(b) with rate between α and $\alpha_{\max}$" is a **conjecture**, and for the chaotic benchmarks even the counterfactual-past step is unavailable (Lemma 11). *The fibre is a needle.*

Writing $Q = \{\int_0^\infty e^{As}PGu\,ds : |u| \le \bar{u}\}$, a symmetric convex body with support function $h_Q(w) = \bar{u}\int_0^\infty |w^\top e^{As}PG|\,ds$, the fibre diameter is computed exactly as $\delta(L) = \max_{|d|=1} 2/g_Q(e^{-AL}Pd)$ with g_Q the gauge of Q (a linear programme on a Riemann discretisation of Q). For the oscillator with $n_x = 2$ the maximiser is the single direction d with $Pd \in S_k$: $\delta(L)$ decays at rate 1.000α ($n_z = 3$) and 2.617α ($n_z = 5$; $\alpha_k = 2.618\alpha$), every other direction decays at $\approx \alpha_{\max}$, and the angular half-width of the needle shrinks from 3° to 1° over $L = 2$–6 (Table 2). An RMS error curve averages over directions, so a sweep's knee on a log scale is expected between the slopes $-\alpha_k$ and $-\alpha_{\max}$, with the slow needle visible only at the largest L. *Tested by* C2-window-a1 (six dev systems, one bandwidth; pre-registered rule "$e(L \ge \ln(1/\eta)/\alpha)$ within 5% of $e(512)$ on $\ge 5/6$": **FAIL**, `id` 2/5, `sequence_ood` 3/5) and C2-window-a2 (battery, duffing, vdp at 2α, the only test of the $1/\alpha$ scaling; rule "knee in samples changes by $\le 2\times$ on $\ge 2/3$": **FAIL**, 1/2). The ℓ-free rule that was pre-registered is the sufficient bound with ℓ dropped; the corollary below restores ℓ and is *post hoc* consistent with the largest failure (battery), not with lorenz (Table 1). Source: `paper/generated/RESULTS.md`.

Corollary 13 (practical window rule, and why one plant needs more)**.** *[proven: (a) of the theorem, rewritten; its use as an explanation of the sweep is* post hoc*, not pre-registered] For a target relative error η the window that is* sufficient *is*

$$L^*(\eta) \;\approx\; \frac{\ln(1/\eta) + \ln \ell}{\alpha} \qquad \textit{(plus the design constant } \ln(\kappa\|B\|\bar{y}/\alpha)/\alpha\textit{)},$$

where ℓ is the Lipschitz constant of the inverse of the windowed map (A4). The $\ln(1/\eta)/\alpha$ part is plant-independent; the plant enters through $\ln \ell$ and, in the design constant, through $\bar{y}$. The lower bound of the theorem (linear plants) can only shorten *the necessary window (rate $\alpha_k \ge \alpha$), so a plant whose sufficient window exceeds $\ln(1/\eta)/\alpha$ does so through ℓ (or $\bar{y}$), not through the fading-memory rate.*

Integrator-type states (coulomb counting; a heuristic reading, not a proven statement). If a state enters the output only weakly, as the battery's state of charge enters the terminal voltage through the open-circuit-voltage slope $\partial V_{\mathrm{oc}}/\partial \mathrm{SOC}$, then the latent carries that state at gain $\propto$ slope and inverting it costs $\ell \propto 1/\text{slope}$: large ℓ, long L^*. The state itself is the integral of the input (coulomb counting), so a decoder must integrate the window, which is possible only over the window's length; the memory needed is $\ln \ell/\alpha$ longer than for a well-observed state. This is the sufficient side of the theorem; the digest's earlier attribution of the battery to "the Lemma 1′ worst case" is withdrawn: Lemma 11(b) is plant-independent (rate exactly α for every plant) and the lower bound at $n_z - n_x + 1 = n_z$ (scalar integrator) gives $\alpha_k = \alpha_{\max}$, the *shortest* necessary window.
Proxy for ℓ. No node measures ℓ itself; the phase-1 diagnostic `l_dec_q95` (random-direction quotient of the *stationary* base decoder, $\approx \|J\|_F/\sqrt{n_z}$, Remark 25) is used as its proxy in Table 1. It is an estimate on data of a related constant, not the constant of the bound.
Reading the table (post hoc). The pre-registered ℓ-free rule fails where ℓ is largest (battery, $1.73\times$ the $L = 512$ error at $L = 256$) and marginally on lorenz (1.07). With $\ln \ell$ restored the rule can be evaluated only on bioreactor, lorenz and vdp: on battery, duffing and quad_tank the refined L^* leaves no grid window below 512, so the refined rule is *untested* there (it does not "hold"); it holds on bioreactor and vdp and still fails by 7% on lorenz (2/3 on `id` and 2/3 on `sequence_ood` in the results layer). For battery the refined rule only says that its failure at $L \ge L_0$ is not in contradiction with the theory. This is what a sufficient-side statement can do: the measured knee sits at or below it (duffing's knee is at 32 samples against $L^* = 281$; the bound is far from tight there), and it does not predict the knee. Bioreactor is non-monotone (longer windows are *worse*: an optimisation or over-fitting effect that no truncation bound describes) and quad_tank's fixed $dt = 0.05$ puts both rules beyond the grid, as predicted below. The refined rule is offered as the

Table 1: Post-hoc reading of the C2-window-a1 sweep (lora, `sequence_ood`, NRMSE at $L \in \{1, \dots, 512\}$ samples, $\eta = 0.05$). $L_0 = \ln(1/\eta)/(\alpha dt)$ is the pre-registered ℓ-free rule and $L^* = (\ln(1/\eta) + \ln \ell)/(\alpha dt)$ the rule of Corollary 13 with $\ell =$ `l_dec_q95` of the node's own phase 1 (both in samples; the design constant is omitted; not pre-registered). "ratio" is $\max_{L \geq L_\cdot} e(L)/e(512)$ on the dyadic grid; the node's criterion is ≤ 1.05; n.t. = not testable (no grid window in $[L_\cdot, 512)$, so the ratio is $e(512)/e(512)$ and says nothing; the results layer leaves these out). ℓ and L^* are the results-layer values. Monotone: no window worse than a shorter one by $> 10\%$.

system	n_z	α	αdt	ℓ	L_0	L^*	ratio, $L \geq L_0$	ratio, $L \geq L^*$	monotone
battery	7	1.5	0.0223	806	135	435	1.73 (fails)	n.t.	yes
bioreactor	5	0.5	0.0309	44	97	219	1.00	1.00	no (min at 128; 2.4× worse at 512)
duffing	5	3.0	0.0309	295	97	281	1.00	n.t.	yes
lorenz	7	2.0	0.0223	10	135	239	1.07 (fails)	1.07 (fails)	yes
quad_tank	6	0.1	0.0050	14	599	1121	n.t.	n.t.	yes (falling at 512)
vdp	5	1.5	0.0309	23	97	199	1.00	1.00	yes

theory's explanation of why the pre-registered test failed, not as a passed test; a pre-registered test of it needs a new node. The $L = 1$ row of the same sweep (current input only, Proposition 14(a)) is +143% to +1522% worse than $L = 512$ on every dev system (`sequence_ood`, lora; paired changes from the C2-window-a1 node report, not a number of the generated results layer).

Proposition 14 (the current input is not enough)**.** *(a) [proven] Under the zero-order hold, $T^0(\cdot, t_k; u)$ does not depend on u_k. (b) [proven, linear SISO plants] For the linear plant of Theorem 12 with $n_y = n_u = 1$ under (H1)–(H2), but without assuming (A, PG) controllable, (z, u_t) determines x iff* $\operatorname{range} P \cap \langle A \,|\, PG \rangle = \{0\}$*, for every $L \geq 0$. If (A, B) is controllable, the eigenvalues of A are distinct and none is a transmission zero of $H(s) = C(sI - F)^{-1}G$, then $\langle A|PG\rangle = \mathbb{R}^{n_z}$ and the condition fails* for every n_z*: no latent dimension makes the latent store the history. (For $n_y > 1$ the condition $p_i^* BH(\lambda_i) \neq 0$ is stronger than "not a transmission zero".) (c) [proven, local] Parametrise the input before $t - L$ by $\theta \in \mathbb{R}^p$ (e.g. held levels in the interior of U), keep the window fixed, and let $\Psi(x, \theta)$ be the resulting latent at t. If Ψ is C^1 near $(\bar{x}, \bar{\theta})$ with trajectories in the interior of $\mathcal{X}$ and* $\operatorname{rank} \partial_\theta \Psi(\bar{x}, \bar{\theta}) = n_z$*, then every x in a neighbourhood of $\bar{x}$ has a past $\theta(x)$ with the same latent and the same window; any decoder $D(z, u_t)$ errs by at least half that neighbourhood's radius.*

Intuition. With a time-varying map the past input is part of what must be inverted, and the latent alone cannot hold it.

Proof. (a) The backward solution on $(-\infty, t_k]$ uses u on $[t_j, t_{j+1})$, $j \leq k - 1$, only. (b) By Theorem 12(b), ambiguity exists iff $Pd = e^{AL}\Delta m$ for some $d \neq 0$ and admissible Δm; the reachable differences form a neighbourhood of 0 in $\langle A|PG\rangle$, which e^{AL} maps onto itself. For distinct eigenvalues, controllability of (A, PG) is $p_i^* PG \neq 0$ for every left eigenvector, and $p_i^* PG = -p_i^* B\, H(\lambda_i)$ (from $p_i^* P(F - \lambda_i I) = p_i^* BC$). (c) Implicit function theorem: choose n_z components of θ with invertible $\partial\Psi$-block and solve $\Psi(x, \theta) = \Psi(\bar{x}, \bar{\theta})$ for them as a C^1 function of x. □

The rank condition in (c) is checkable on a simulator: for Van der Pol with the saturating actuator, twelve held levels before the window give $\operatorname{rank} \partial_\theta \Psi = 5 = n_z$ (smallest singular value $1.6 \cdot 10^{-4}$: the check shows existence of the ambiguity, not its size, which is $\sigma_{\min}\times$ the admissible range of θ). Example: the oscillator $\dot{x}_1 = \omega_0 x_2$, $\dot{x}_2 = -\omega_0 x_1 + u$, $y = x_1$, $n_z = n_x = 2$, with $v \equiv 0$, $u = a\mathbf{1}_{(-\infty, t-L)}$: x and $x' = x - P^{-1}e^{AL}A^{-1}PGa$ share latent, window and current input. On the stationary route the current input does suffice (the injection $\Phi(z, u)$ depends on $u(t)$ only; Section 1); the price there is a small-gain margin (Proposition 7). *Tested by* the $L = 1$ row of

C2-window-a1 (above). No current node compares the context encoders (current, window, GRU, SFM) at a matched budget; C1-ctx only moves the SFM's parameters into the budget and does not test this proposition.

2.1 Predictions for the window sweeps

With $dt = 0.1\sin(\pi/2n_z)/\alpha$, $\alpha\,dt$ does not depend on the bandwidth scale s, and a window of L samples covers $(L-1)\,dt$ seconds of relevant history (Proposition 14a).

1. The knee in seconds scales as $1/(s\alpha)$; in *samples* it is the same for all s (vdp, duffing, battery in C2-window-a2). One e-fold of the rate-α bound is $1/(\alpha dt) = 32.4$ samples ($n_z = 5$) or 44.9 ($n_z = 7$); at rate α_k it is α_k/α times fewer (2.6 at $n_z = 5$, $n_x = 2$; 4.0 for lorenz, $n_z = 7$, $n_x = 3$; the `a2/a` column of `c2_predictions` is the second-slowest mode group, which equals α_k only when $n_z - n_x + 1 \le 4$). The worst-case window of Theorem 12(a) at $\eta = 10^{-3} z_{\text{scale}}$ (ℓ omitted) is $\approx 311, 313, 320, 467$ samples; measured tails are 10^2–$5\cdot10^2$ times below the bound, i.e. the bound overstates the window by $\ln(485)/(\alpha\,dt) \approx 200$ samples. At one bandwidth (C2-window-a1) the knees within 5% of $e(512)$ are 32 (vdp, `id` and `sequence_ood`; duffing and bioreactor on `sequence_ood`), 64 (bioreactor `id`), 128 (quad_tank `id`) and not reached on the grid for battery and lorenz (both tiers), duffing (`id`) and quad_tank (`sequence_ood`); see Table 1. At 2α (C2-window-a2) the knee in samples moved by $\le 2\times$ on only 1/2 evaluable systems: the $1/\alpha$ scaling is *not* supported by the current data.

2. quad_tank has a fixed $dt = 0.05$: $\alpha dt = 0.0025, 0.005, 0.01$ for $s = \frac{1}{2}, 1, 2$; the knee in samples scales as $1/s$ and the grid ($L \le 512$) spans only $1.3, 2.6, 5.1$ e-folds, so the curve should still fall at $L = 512$ for $s \le 1$ (it does at $s = 1$): this plant cannot test the scaling.

3. On a dyadic grid "knee$\times\alpha$ varies by less than $2\times$" holds only if the knee sits at the same grid point for every s (neighbours differ by exactly 2); use "$\le 2\times$" or a knee interpolated in $\log L$.

2.2 Context encoders: forgetting by construction

Lemma 15 (SFM context)**.** *[proven] Let $\xi_{k+1} = A_d^\xi \xi_k + B_d u_k/u_{\text{ref}}$, $c_k = \psi([\xi_k, u_k/u_{\text{ref}}])$, with A^ξ block-normal, $s_b = -\text{softplus}(\cdot) < 0$, $\alpha_\xi = \min_b |s_b|$, and ψ ℓ_ψ-Lipschitz. Then for $l \ge 1$*

$$\textit{(a)}\ \Big\|\frac{\partial c_k}{\partial u_{k-l}}\Big\| \le \frac{\ell_\psi}{u_{\text{ref}}}\|B_d\| e^{-\alpha_\xi (l-1)dt}, \qquad \textit{(b)}\ \ell_E \le \frac{\ell_\psi}{u_{\text{ref}}}\Big(1 + \frac{\|B_d\|}{1 - e^{-\alpha_\xi dt}}\Big),$$

*and the per-lag bound is attained (ψ isometric on the slowest block, B_d in that block). (c) (*the memory probe*) If two input records agree on the last L samples and differ arbitrarily before, with $|u| \le \bar{u}$,*

$$\|c_k - c_k'\| \le \frac{\ell_\psi \|B_d\|\, 2\bar{u}}{u_{\text{ref}}} \frac{e^{-\alpha_\xi L\, dt}}{1 - e^{-\alpha_\xi dt}}.$$

Proof. $\xi_k = \sum_{l\ge1} (A_d^\xi)^{l-1} B_d u_{k-l}/u_{\text{ref}}$, $\|(A_d^\xi)^{l-1}\| = e^{-\alpha_\xi(l-1)dt}$; chain rule. For ℓ_E add the direct path u_k and sum the geometric series. (c) Only lags $l \ge L+1$ differ; sum (a) over them. □

The rate exists for every parameter value but is not bounded below uniformly (softplus$\to 0$); a floor $s_b \le -\alpha_{\min}$ makes it uniform (the C2-floor nodes set $\alpha_\xi \ge c\,\alpha$, $c \in \{0.25, 0.5, 1\}$).

What the probe measures. The C1-where and C2-floor memory probes keep the last L inputs, swap everything older, and fit a rate to the decoded change $|\Delta\hat{x}| \le \ell_c \|c - c'\|$. This is the quantity of (c): a test of the *encoder's* memory, not of Lemma 11 (the latent z is unchanged by the probe).

Since (c) is an upper envelope, a fitted rate below α_ξ over a finite L range does not contradict it unless the curve is in its exponential regime over several decades; the probe tests the rate only there.

At initialisation the SFM's fill-in poles are $\alpha/1.5^b$, $b = 1, \dots, 5$, so its slowest mode is $\alpha/1.5^5 = 0.132\alpha$ and the next 0.198α: at initialisation the SFM forgets $\approx 7.6\times$ more slowly than the latent. *Measured (un-floored, C1-where, lora, median over seeds;* `paper/generated/RESULTS.md`*):* the learned SFM's slowest pole is 0.14–0.18 α (battery 0.18, bioreactor 0.14, duffing 0.14, lorenz 0.18, quad_tank 0.17, vdp 0.15), i.e. between the two slowest *initial* poles: training moves the slowest pole only slightly (by at most $\approx 40\%$, never past the second initial pole), so the "5–7× extra memory" of the un-floored SFM is largely inherited from initialisation rather than learned. *The floor intervention (C2-floor-025/05/1) fails its pre-registered rule.* At 0.5α: no significant loss on 4/6 dev systems (rule: $\geq 5/6$; battery +23–36% and bioreactor lose), TOST non-inferiority at $\pm 10\%$ on 0/6, and the strict probe check "fitted rate $\geq$ floor" on 0/6 (ratios 0.73–0.96: battery 0.78, bioreactor 0.77, duffing 0.73, quad_tank 0.96, vdp 0.82; lorenz −0.02, flat). At 0.25α: no significant loss on 6/6 but TOST 3/6 and probe 1/6; at 1α: 4/6, 0/6, 0/6 (battery $\approx 2\times$ worse). So the data are *consistent with* the memory being unused on four systems but do not establish it (three seeds; see the TOST counts), and battery, the large-ℓ plant of Corollary 13, is the one system where a floor at 0.5α clearly costs accuracy, as the corollary says it should (its sufficient window $\ln(1/\eta)/\alpha + \ln \ell/\alpha$ is attenuated by $e^{-0.5\alpha L^*}$ under the floor); the bioreactor loss has no such explanation. The probe ratios below one do not contradict Lemma 15(c), which is an upper envelope, but they mean the pre-registered check of its *rate* fails on every system, and the flat lorenz curve ($d(L)$ within ×2–3 over $L \leq 5.7$ s) is unexplained: a probe outside its exponential regime, to be reported as such. For the window prediction the SFM has no truncation loss on the latent's time scale (it should match the window encoder's plateau beyond the knee, not its knee), but an input glitch persists 7.6× longer in c and ℓ_E is correspondingly larger ($\|B_d\|/(1 - e^{-\alpha_\xi dt}) \approx 7.6/(\alpha dt) \approx 246\|B_d\|$ for $n_z = 5$).

Remark 16 (SFM is an exact context for linear plants)**.** If $\mathrm{spec}(A^\xi) \supseteq \mathrm{spec}(A)$ with nonzero input weights (true at initialisation), then $q(t_k) = -\sum_{i<k} A_d^{k-1-i} A^{-1}(A_d - I)PGu_i$ is a fixed linear function of ξ_k (each 2×2 block commutes with J): no truncation, whereas a window of L samples truncates at $e^{-\alpha(L-1)dt}$. Once the poles are trained this holds only approximately.

Remark 17 (a GRU has no such guarantee)**.** (1) An update-gate bias b gives $h_k = \sigma(b)h_{k-1} + \dots$, forgetting at rate $-\ln \sigma(b) \to 0$ as $b \to \infty$. (2) No fading memory in the sense of Boyd and Chua [4]: with reset gate $\equiv 1$ and update gate $\equiv 0$, $h_{k+1} = \tanh(3h_k + 2u_k)$ is bistable; one pulse flips the state and the context difference stays 1.99 for all later k, although the two inputs agree after the pulse. Contractivity of recurrent networks needs weight constraints [12]; the SFM has it by construction. *Tested by* no current node (an encoder ablation SFM vs GRU at a matched budget is requested; C1-ctx does not contain a GRU variant).

3 Error bound for the deployed observer

Setting. Plant $\dot{x} = f(x, u)$, $y = h(x)$, input held on $[t_k, t_{k+1})$, $t_k = k\,dt$. The decoder-side observer, exactly as deployed, is

$$\hat{z}_{k+1} = A_d \hat{z}_k + W_0 \tilde{y}_k + W_1 \tilde{y}_{k+1}, \qquad \hat{x}_k = \hat{T}^+(\hat{z}_k; \hat{c}_k), \qquad \hat{c}_k = E(\tilde{u}_0, \dots, \tilde{u}_k), \tag{2}$$

with A_d, W_0, W_1 the exact step of Lemma 26(a), the context held between samples, and $\hat{z}_0$ a fixed function of the prior $\hat{x}_0$. Its response to a sensor sequence (v_k) is the *latent noise filter*, the sampled

Table 2: Numerical checks of this section (CPU).

result	check	outcome
Lemma 11	integrator; oscillator $n_z = 5$; Van der Pol	rate $1.5000 = \alpha$; $1.0000 = \alpha$ (both); bound loose 135–485×
Thm. 12(b)	$\delta(L)$ by gauge/LP, oscillator $n_x = 2$	rate $1.000, 2.617$ at $n_z = 3, 5$ ($\alpha_k = 1, 2.618$); needle at predicted angle
	bounds $r_q e^{-\alpha_k L}/\|P\| \le \delta \le 2R_q e^{-\alpha L}/\sigma_{\min}$	hold at every L (slack 21×, 1.5–106×)
	$\sigma_{\min}(e^{-AL}P)$ (arithmetic)	same rates $1.000, 2.618$
Cor. 13	Table 1 from the C2-window-a1 report	ℓ-free rule fails 1.73× (battery); with $\ln \ell$: holds on 2/3 testable systems (lorenz 1.07), untestable on battery, duffing, quad_tank
Prop. 14(c)	Van der Pol, 12 past levels	rank $\partial_\theta \Psi = 5$
sampled form	ZOH step identity; discrete constant	exact to 10^{-15}; $c_d = 0.985 \le 1/\alpha$
Lemma 15(a,b)	tight SFM; random SFM	ratio 1.0000000000; ≤ 0.13
Lemma 15(c)	history swap, floor 0 and 0.5; initial poles	observed/bound ≤ 0.01; $\alpha_\xi = \alpha/1.5^5$ at init, $\ge 0.5\alpha$ with floor
GRU remark	latch in the code's GRU encoder	difference 1.990 at $k = 100, 1000, 4999$

form of $(sI - A)^{-1}B$:

$$H_0 = W_1, \qquad H_m = A_d^{m-1}W_0 + A_d^m W_1 \ (m \ge 1), \qquad \gamma_y := \sum_{m\ge 0} \|H_m\| \le \frac{\kappa\|B\|}{\alpha}, \qquad \|H\|_2^2 := \sum_{m\ge 0} \|H_m\|_F^2, \tag{3}$$

where the inequality holds at every dt (Lemma 26(c)).

Reference condition. Every bound is stated relative to the *nominal* deployed observer $\hat{x}_k^{\text{nom}}$: the same episode, the nominal sensor model (the one used in training, `data.sensor`), the true input record and the nominal prior. Deviations are $v_k := \tilde{y}_k - \tilde{y}_k^{\text{nom}}$ (sensor faults, extra noise), $\hat{c}_k - c_k^{\text{nom}}$ (input corruption) and $\hat{z}_0 - \hat{z}_0^{\text{nom}}$ (initialisation). This is the condition the decoder was trained in: phase 2 regresses x_k on the deployed latent of the nominal-noise record, so its validation error measures the nominal error directly, and no noise-free quantity is needed.

Assumptions. (A1)–(A4) and the regularity clause ($f(\cdot, u) \in C^1$, $h \in C^2$, hence $M_2 := \sup \|\ddot{y}\|$, $L_y := \sup \|\dot{y}\|$ on each hold interval) of Assumption 1, and

(L) $\hat{T}^+ \in C^1$, and the context encoder satisfies $\|E(\tilde{u}_{0:k}) - E(u_{0:k})\| \le \ell_E \sup_{j\le k} \|\tilde{u}_j - u_j\|$ (for SFM, $\ell_E \le (\ell_\psi/u_{\text{ref}})(1 + \|B_d\|/(1 - e^{-\alpha_\xi dt}))$, Lemma 15).

Write ℓ_z, ℓ_c for the sup of $\|\partial_z \hat{T}^+\|$, $\|\partial_c \hat{T}^+\|$ over the segments a given episode traverses (defined in Proposition 19), and $\ell_z^{\text{g}}, \ell_c^{\text{g}}$ for the global constants (finite for tanh/SiLU networks on a bounded context set, for every conditioner; for FiLM, LoRA and hyper the available bound is a product of layer norms that is orders of magnitude larger, Proposition 7(b)). The latent deviations are linear, input-free and decoder-independent, so they are fixed before any Lipschitz constant is used.

Definition 18 (nominal error, held-out quantile)**.** For an episode $\omega \sim \mathcal{D}$ (initial state and input path from the training episode distribution) let $e_{\text{nom}}(\omega) := \sup_{k\in S} \|\hat{x}_k^{\text{nom}} - x(t_k)\|$, S the steady window, and let ε^β be its $(1-\beta)$-quantile. For n i.i.d. held-out episodes, $\hat{\varepsilon} := \max_i e_{\text{nom}}(\omega_i)$ satisfies

$$\mathbb{P}\big(\hat{\varepsilon} \ge \varepsilon^\beta\big) \ge 1 - (1-\beta)^n \qquad (n = 64,\ \beta = 0.05 :\ 0.96), \tag{4}$$

because all n draws fall below the quantile with probability at most $(1-\beta)^n$.

ε^β contains everything the nominal observer gets wrong: approximation, window truncation (by Lemma 11(a) and Theorem 12(a) at most $\frac{\ell}{2}\varepsilon(L)$ for a sufficient context), nominal noise and the discretisation (Lemma 26(b) only relates the deployed latent to the continuous map, $\|\bar{z}_k - z(t_k)\| \le e_{\mathrm{disc}} = O(dt^2)$). It is a quantile with distribution-free confidence, not a supremum, and only for $\mathcal{D}$. It is defined on the *true* plant's episodes (phase 2 regresses true-plant state labels on the deployed latent): no model of f enters it, which is the point of Proposition 23.

Proposition 19 (decoder-side observer: ISS relative to the nominal observer)**.** *[(i), (ii), (iv) proven and exact; (iii) proven as a decomposition, see the paragraph after the proof] Assume (A1)–(A4), (L). Let $\Delta z_k := \hat{z}_k - \hat{z}_k^{\mathrm{nom}}$, $\Delta c_k := \hat{c}_k - c_k$.*

(i) (latent deviation, exact) $\Delta z_k = A_d^k \Delta z_0 + \sum_{i\le k} G_i^{(k)} v_i$ with $G_i^{(k)} = H_{k-i}$ $(i \ge 1)$, $G_0^{(k)} = A_d^{k-1} W_0$; hence $\|\Delta z_k\| \le \kappa e^{-\alpha t_k}\|\Delta z_0\| + \frac{\kappa\|B\|}{\alpha}\sup_{i\le k}\|v_i\|$ for every k, and $\limsup_k \|\Delta z_k\| \le \gamma_y \limsup_k \|v_k\|$.

(ii) (path identity, exact)

$$\hat{x}_k - x(t_k) = \big(\hat{x}_k^{\mathrm{nom}} - x(t_k)\big) + \bar{J}_{z,k}\Delta z_k + \bar{J}_{c,k}\Delta c_k, \tag{5}$$

$\bar{J}_{z,k} = \int_0^1 \partial_z \hat{T}^+(\hat{z}_k^{\mathrm{nom}} + s\Delta z_k; \hat{c}_k)ds$, $\bar{J}_{c,k} = \int_0^1 \partial_c \hat{T}^+(\hat{z}_k^{\mathrm{nom}}; c_k + s\Delta c_k)ds$.

(iii) (ISS bound) If $\|v_k\| \le \delta_y$ and $\|\tilde{u}_k - u_k\| \le \delta_u$, then for every k

$$\|\hat{x}_k - x(t_k)\| \le \|\hat{x}_k^{\mathrm{nom}} - x(t_k)\| + \ell_z\Big(\kappa e^{-\alpha t_k}\|\Delta z_0\| + \frac{\kappa\|B\|}{\alpha}\delta_y\Big) + \ell_c\,\ell_E\,\delta_u, \tag{6}$$

with ℓ_z, ℓ_c the sup of the Jacobian norms on the traversed segments $[\hat{z}_k^{\mathrm{nom}}, \hat{z}_k]\times\{\hat{c}_k\}$ and $\{\hat{z}_k^{\mathrm{nom}}\}\times[c_k, \hat{c}_k]$; on the steady window the first term is $\le \varepsilon^\beta$ for a $(1-\beta)$ fraction of episodes, and in the limsup $\kappa\|B\|/\alpha$ may be replaced by the exact γ_y.

(iv) (stochastic deviations) For zero-mean i.i.d. v_k with covariance $\sigma_v^2 I$ (e.g. extra Gaussian noise), $\limsup_k(\mathbb{E}\|\hat{x}_k - x(t_k)\|^2)^{1/2} \le \limsup_k(\mathbb{E}\|\hat{x}_k^{\mathrm{nom}} - x(t_k)\|^2)^{1/2} + \ell_z^{\mathrm{g}}\sigma_v\|H\|_2 + \ell_c\ell_E\delta_u$, $\|H\|_2^2 = dt\,\mathrm{tr}W_c + o(dt)$, $AW_c + W_cA^\top + BB^\top = 0$. The global constant is needed because unbounded noise leaves every neighbourhood of the data with positive probability.

Intuition. The learned map never sees raw measurements: sensor deviations and initialisation reach it through a known, input-free stable filter, a wrong input record moves only the context, and each deviation is multiplied by the decoder's slope along the path it actually travels.

Proof. (i) $\hat{z}$ and $\hat{z}^{\mathrm{nom}}$ obey the same linear recursion with forcing difference $W_0v_k + W_1v_{k+1}$; unroll it, and use $\|A_d^k\| \le \kappa e^{-\alpha t_k}$ and $\sum_i \|G_i^{(k)}\| \le \kappa\|B\|/\alpha$ (Lemma 26(c)). (ii) Split $\hat{T}^+(\hat{z};\hat{c}) - \hat{T}^+(\hat{z}^{\mathrm{nom}}; c)$ at $\hat{T}^+(\hat{z}^{\mathrm{nom}};\hat{c})$ and apply the fundamental theorem of calculus on each segment. (iii) Norms in (5), (i), (L), Definition 18. (iv) $L^2(\mathbb{P})$ norms of the pointwise inequality (Minkowski) with $\|\bar{J}_z\| \le \ell_z^{\mathrm{g}}$; $\mathbb{E}\|\sum_i G_i^{(k)}v_i\|^2 = \sigma_v^2\sum_i\|G_i^{(k)}\|_F^2 \to \sigma_v^2\|H\|_2^2$, a Riemann sum ($H_m = e^{At_m}B\,dt + O(dt^2)$) of $dt\int_0^\infty \|e^{As}B\|_F^2 ds$. □

When the nominal sensor is noise-free, (6) is the classical decomposition: fit error plus decoder slope times the ISS gain $\kappa\|B\|/\alpha$ of the latent filter [cf. 13, Lemma 17, Theorem 19], here for the sampled observer and with an input-corruption term. The latent error equation of Shaaban et al. [16, Proposition 1] assumes no injection; with injection $\Phi \ne 0$ the latent deviation is no longer

input-free and the margin of Proposition 7 is needed (sampled form: Lemma 26(d)). The bound has no amplitude-dependent condition, but its constants ℓ_z, ℓ_c may depend on the context.

(iii) is a decomposition with measured constants, not a certificate. With ℓ_z, ℓ_c the suprema over the traversed segments, (iii) is (ii) with norms taken, and the constants are only known after the fact (lower bounds of the suprema when estimated from samples). Evaluated *a priori*, with the phase-1 proxy `l_dec_q95` for the global constant ℓ_z^{g} and the nominal sensor level $\delta_y = 0.03\, y_{\mathrm{std}}$, the second term alone is $\ell_z \kappa \|B\| \delta_y / \alpha = 66$ (battery; $\kappa\|B\|/\alpha = 2.68$, $\ell = 816$), 15 (quad_tank; 37.4, 13.5), 6.7 (duffing), 5.9 (bioreactor), 1.1 (vdp), 0.40 (lorenz), against measured nominal NRMSE 0.016–0.052 (C1-where, lora, `id`): 15× to 1400× too large (`check_apriori_vacuity`). The bound is therefore vacuous as a guarantee on every benchmark and is *not* presented as one. What it offers is the exact attribution (ii) and the exact stochastic form (iv), both of which are measurable (Definition 24).

Remark 20 (reading the bound)**.** Three quantities set the error: the nominal error ε^β (capacity, data, window), the decoder slope ℓ_z times a design-only latent deviation ($\propto 1/\alpha$), and the input sensitivity $\ell_c \ell_E$, which is zero for the autonomous observer. The architecture enters only through these constants: the bound says which one a conditioning mechanism must keep small, not which mechanism wins.

What would test it (eval-only; implemented by no current node). Under `input_noise/bias/delay` the excess error is $\bar{J}_c \Delta c$ exactly: the autonomous observer is unaffected, and variants would order by their realised input gain $\|\bar{J}_c \Delta c\| / \|\Delta c\|$ if that gain were measured on saved checkpoints (Definition 24; C3-faults has the conditions, no node records the gains). Under `poor_init` the excess is $\bar{J}_z A_d^k \Delta z_0$, gone after a few $1/\alpha$ (C3-faults records the convergence time, which is the observable consequence). Upper bounds do not rank errors, so none of this is implied by the bound.

Check (`m3_checks.py::check_noise_iss`, `check_input_corruption`, `check_identity`; linear plant with exact maps $z = Px + \nu$, decoder $P^+(z - c)$, $\ell_z = \ell_c = 12.48$, $\ell_E = 0.048$, $n_z = 5$, $\alpha = 1.5$, $dt = 0.0206$ as for vdp). Worst-case bounded deviation: latent $0.0372 \le \gamma_y \delta_y = 0.0537 \le \kappa\|B\|\delta_y/\alpha = 0.0745$. White deviation, RMS latent 0.00427 (Monte Carlo) vs $\sigma_v \|H\|_2 = 0.00430$; the sup form at $\delta_y = 4\sigma_v$ is 30× looser, so only (iv) is informative for Gaussian conditions. Input bias 0.1 / delay 5 / bounded noise: state error $0.022 \le 0.061$, $0.197 \le 1.21$, $0.024 \le 0.18$. Identity (5) for a tanh-MLP decoder holds to 4×10^{-10} (finite-difference Jacobians); its realised gain differs from the on-data $\|\partial_z \hat{T}^+\|_2$ by a factor 0.01–1.38 (median 0.32), so on-data slopes neither bound nor estimate it. Coverage of (4) at $n = 64$, $\beta = 0.05$: 0.961 vs 0.963.

Corollary 21 (sensor faults: bounded and forgotten, uniformly in the state)**.** *For a decoder-side observer and any sensor deviation v, pathwise $\|\hat{x}_k - x(t_k)\| \le \|\hat{x}_k^{\mathrm{nom}} - x(t_k)\| + \ell_z^{\mathrm{g}} \|\sum_i G_i^{(k)} v_i\|$. In particular: an outlier of size M at k_0 adds at most $\ell_z^{\mathrm{g}} \kappa \|B\| M \left(w_1 + e^{\alpha dt} w_0\right) e^{-\alpha(t_k - t_{k_0})} \le \ell_z^{\mathrm{g}} \kappa \|B\| M\, dt\, e^{\alpha dt} e^{-\alpha(t_k - t_{k_0})}$ (w_0, w_1 of Lemma 26(c), $w_0 + w_1 \le dt$), with time integral at most $\ell_z^{\mathrm{g}} \kappa \|B\| M\, dt/\alpha$, for every state, input and M; an outlier process (each sample with probability p, size M, random sign) adds at most $\ell_z^{\mathrm{g}} \gamma_y p M$ in mean and $\ell_z^{\mathrm{g}} \sqrt{p} M \|H\|_2$ in RMS; sample-and-hold dropout adds at most $\ell_z^{\mathrm{g}} \gamma_y L_y\, dt\, p/(1-p)$ in mean; a drift b adds at most $\ell_z^{\mathrm{g}} \gamma_y \sup \|b\|$, and for slow drift $\Delta z_k \approx -A^{-1} B b_k$, since the discrete DC gain is $(I - A_d)^{-1}(W_0 + W_1) = -A^{-1}B$ exactly; coloured deviations obey $\mathbb{E}\|\Delta z\|^2 = \sum_{m,m'} \mathrm{tr}(H_m R_v (m' - m) H_{m'}^\top)$.*

Intuition. There is no feedback from the estimate to the latent: a fault is a forcing of a fixed stable filter, so the observer's own error can never amplify it.

Proof. Proposition 19(ii) with $\Delta c = 0$, $\Delta z_0 = 0$, $\|\bar{J}_z\| \le \ell_z^{\mathrm{g}}$. Outlier: $\Delta z_{k_0} = W_1 M e$ and $\Delta z_k = (A_d^{k-k_0-1} W_0 + A_d^{k-k_0} W_1) M e$, bounded with the merged kernels of Lemma 26(c); the integral is

$dt \sum_m \|H_m\| M \le dt\,\kappa \|B\| M/\alpha$. Process: $\mathbb{E}\|v_i\| \le pM$ in (i), and $R_v(m) = pM^2\delta_m$. Dropout: a held sample differs from the current one by at most $L_y\,dt\,N_k$, N_k geometric with $\mathbb{E}N_k \le p/(1-p)$. DC gain: $W_0 + W_1 = A^{-1}(A_d - I)B$ and A, A_d commute; the quasi-static error is bounded by summation by parts, $\|\Delta z_k + A^{-1}Bb_k\| \le \sum_{m\ge 1} \|\sum_{j\ge m} H_j\| \sup_j \|b_j - b_{j-1}\|$. □

What this proves about filters, and what it does not. The corollary is a *qualitative* ISS guarantee: the excess is bounded and decays at rate α for every state, input and fault size, given ℓ_z^{g}, with no feedback from the estimate to the latent. The EKF's stochastic stability is local (small initial error and noise, 15) and an outlier moves its linearisation point. A qualitative guarantee is not a comparative advantage, and the data say so: on *exact* models no C3-faults condition (sensor, input, plant-drift and outage) reaches the learned-wins regime of $\ge 4/6$ systems (learned-wins regime: none of 15 conditions; the learned observer beats the best filter on 0/6 under every sensor-side condition and on at most 2/6 against the best filter of either kind under input bias, input delay or 10% plant drift), and it wins on 0/6 under colored Student-t noise (C3-noise-ct), as Remark 22 predicts for linear plants and as the EKF's 2.7–14.1× (median 5.9×) smaller exact-model `sequence_ood` error on the dev systems (C1-where) makes expected. Worse, the best *learned* observer is itself above climatology (a failed estimator) under exact-model `outliers` (battery, duffing, quad_tank) and under 30% post-training plant drift (bioreactor, quad_tank): the corollary bounds the excess by $\ell_z^{\mathrm{g}}\times$(filter response), and with ℓ_z^{g} in the hundreds that bound is consistent with failure. Where learned observers win is model incompleteness (C3-mismatch10/20/30/50, C3-partial, C3-realistic), and the theoretical lever there is not robustness to sensor faults but the model term of Proposition 23.

Exact filter-level ratios (design-invariant). For the Butterworth design with $dt = 0.1/\|A\|$, the *latent* deviations of the stress conditions, relative to the nominal noise ($0.03\,y_{\mathrm{std}}$), do not depend on the system: RMS ×3.4 for `colored` ($\varrho = 0.95$), ×57 for `outliers` ($p = 0.02$, $M = 12\,y_{\mathrm{std}}$), and ×30–32 at the quasi-static peak for `sensor_drift` ($0.3\,y_{\mathrm{std}}$); `dropout` costs $O(L_y dt)$ per held sample. These are exact properties of the known filter; the *state*-level excess multiplies them by the variant's realised $\|\bar{J}_z\|$, which no current node records (Definition 24). *Check* `m3_checks.py::check_fault_ratios` computes these ratios for vdp, duffing, bioreactor and lorenz (all agree to two digits; drift $\times 32.1, 32.1, 32.1, 30.5$). `check_outliers`, `check_dropout_drift`: outlier peak $0.519 \le 0.536$, envelope never violated, integral $0.266 \le 0.369$; process mean $0.174 \le 0.258$, RMS 0.241 vs 0.243; dropout ($p = 0.1$) mean $0.00105 \le 0.00737$; DC identity to 10^{-16}; random-walk drift peak $0.137 \le 0.322$, quasi-static deviation $0.0052 \le 0.049$.

Remark 22 (Kalman filter vs. learned KKL on linear plants)**.** For $\dot{x} = Fx + Gu$, $y = Cx$ the exact maps are $z = Px + \nu$ with $PF = AP + BC$, $\dot{\nu} = A\nu - PGu$ and decoder $P^+(z - \nu)$, so the KKL estimate is an LTI filter of (y, u) with sensor-to-error response $\Gamma_m = P^+ H_m$; for $n_z = n_x$ it is the Luenberger observer with gain $P^{-1}B$. The steady KF has the same structure with gain K. (1) Gaussian noise with correct (Q, R): the KF is the MMSE estimator, so no KKL design beats it. (2) White non-Gaussian noise (Gaussian plus outliers): every LTI filter has steady MSE $\mathrm{MSE}_w + (\sigma_v^2 + pM^2)\|\Gamma\|_{H_2}^2$, the KF with $R = \sigma_v^2 + pM^2$ is the best linear filter [1], and the KKL observer against a KF tuned to the nominal R is decided by $\|\Gamma\|_{H_2}$ alone. *Check* (`check_kf_comparison`; oscillator, $\sigma_v = 0.03$, $p = 0.02$, $M = 12$): Gaussian MSE KF $1.98 \times 10^{-3} <$ KKL 6.75×10^{-3}; with outliers KF(nominal R) 0.514, KF(R_{eff}) 0.0115, KKL 2.40, each within 2% of the formula. The KKL loses to both here ($\|P^+\| = 12.5$): outlier robustness is not a general property of KKL observers, and on the benchmarks any advantage over the EKF must be attributed to its mis-tuning, local stability and linearisation, or, as the data show, to the filter's model being wrong (next).

3.1 Model incompleteness: the term that decides C3

Proposition 23 (persistent model error: filter bias floor vs. learned fit error; linear plants)**.** *[proven for linear plants] Let the true plant be $x_{k+1} = Fx_k + Gu_k + d_k + w_k$, $y_k = Cx_k + v_k$, where d_k is the part of the dynamics the modeller's model lacks (a wrong F or a missing term: $d_k = \Delta F\, x_k + \Delta G\, u_k$), w_k, v_k white with covariances Q, R. Any fixed-gain predict–correct observer built on the model (F, G, C), in particular the steady-state Kalman filter and every Luenberger observer, $\hat{x}_{k+1} = \hat{x}^-_{k+1} + K(y_{k+1} - C\hat{x}^-_{k+1})$, $\hat{x}^-_{k+1} = F\hat{x}_k + Gu_k$, has error $e_k = x_k - \hat{x}_k$ obeying*

$$e_{k+1} = (I - KC)F\, e_k + (I - KC)(d_k + w_k) - Kv_{k+1}.$$

*(i) (*bias*) For a constant $d_k \equiv \bar{d}$ (e.g. a constant input and $\Delta F\bar{x}$ at the steady state), the error converges to the non-decaying bias $\bar{e} = \big(I - (I - KC)F\big)^{-1}(I - KC)\,\bar{d}$; in continuous time, for $\dot{e} = (F - KC)e + \Delta F\, x$, $\bar{e} = -(F - KC)^{-1}\Delta F\,\bar{x}$, the DC value of $(sI - (F - KC))^{-1}\Delta F\, x$.*

*(ii) (*bias–variance floor*) Let $Q \succ 0$, $R \succ 0$. For a stationary d_k independent of (w, v) (e.g. d evaluated along the noise-free trajectory, as in the check; with $d_k = \Delta F\, x_k$ along the noisy trajectory the part $\Delta F\, x^{\mathrm{sto}}_k$ is an extra process noise correlated with w, and Σ must then be taken from the joint Lyapunov equation of (x^{sto}, e)) the steady mean-square error splits exactly as $\mathrm{MSE}(K) = b^2(K) + \operatorname{tr}\Sigma(K)$, with b^2 the mean square of the deterministic response to d through $\Phi_K = (I - KC)F$ and $\Sigma = \Phi_K\Sigma\Phi_K^\top + (I - KC)Q(I - KC)^\top + KRK^\top$. Inflating the process covariance to $Q + \lambda I$ (the oracle filters' `q_inflate`) moves $K(\lambda)$ along a one-parameter curve that trades b^2 against $\operatorname{tr}\Sigma$ (in the check: $\|K\|$ and $\operatorname{tr}\Sigma$ increase, b^2 decreases, monotonically); the best achievable error $\mathrm{MSE}^\star := \inf_K \mathrm{MSE}(K) \geq \inf_K \operatorname{tr}\Sigma(K) > 0$ depends on $(F, G, C, \Delta F, \Delta G, Q, R)$ and the input, and on no data. Only n_y directions of d can be cancelled by K, so b^2 keeps a positive limit as $\lambda \to \infty$ whenever d has a component K cannot cancel (e.g. $C\bar{d} = 0$, as for the missing stiffness term of the check).*

*(iii) (*learned decoder-side observer*) Its error obeys Proposition 19 with ε^β measured on the true plant: $\|\hat{x}_k - x_k\| \leq \varepsilon^\beta + \ell_z\|\Delta z_k\| + \ell_c\|\Delta c_k\|$ on a $(1 - \beta)$ fraction of episodes, and no term depends on the model. In the linear case, a decoder fitted on true-plant data is $P^+_{\mathrm{true}}(z - \nu_{\mathrm{true}})$ with $P_{\mathrm{true}}F_{\mathrm{true}} = AP_{\mathrm{true}} + BC$ (continuous-time Sylvester equation of the true plant; for the sampled plant its discrete counterpart): its error is driven by w, v only (no bias), with a mean-square error $\mathrm{MSE}_{\mathrm{KKL}}$ that is a property of the true plant and the noise; at a fixed true plant it does not depend on the modeller's model at all. The model enters the learned observer only through the phase-1 PDE regulariser of T_0 (and through `kkl_mb`, which is a model-based filter and belongs to the other class).*

Prediction. *The learned observer beats the best (oracle-tuned) filter iff $\mathrm{MSE}^\star > \varepsilon^2$, where ε is its fit error on the true plant. For a missing term $s\,d$ of fixed shape and scale s (at a fixed true plant), $\mathrm{MSE}^\star(s) = \inf_K[s^2 b_1^2(K) + \operatorname{tr}\Sigma(K)]$ (b_1^2 the bias term at $s = 1$; d independent of (w, v) as in (ii)) is nondecreasing in s and ε does not depend on s, so there is at most one crossing: filters win below a threshold in the model error (possibly 0) and learned observers above it. The threshold depends on the plant and the noise.*

Intuition. A model-based filter corrects its wrong model only through the n_y measured directions and only at the cost of trusting the noisy measurement more; a decoder trained on the true plant has no model to be wrong about, it pays its fit error instead.

Proof. Subtract the predict–correct recursion from the true one: $x_{k+1} - \hat{x}^-_{k+1} = Fe_k + d_k + w_k$ and $e_{k+1} = (I - KC)(x_{k+1} - \hat{x}^-_{k+1}) - Kv_{k+1}$. (i) Set $e_{k+1} = e_k$ with $w = v = 0$; $(I - (I - KC)F)$

is invertible when the filter is stable. Continuous time: set $\dot{e} = 0$. (ii) Linearity: the response to d (deterministic) and to (w, v) (zero mean, independent of d) are orthogonal in L^2, so the mean squares add; Σ is the stationary covariance of the noise response (discrete Lyapunov equation). $\operatorname{tr}\Sigma(K)$ is the error of the linear filter K under the nominal model and the true noises, so it is bounded below by the nominal-model MMSE, which is positive for $Q \succ 0$, $R \succ 0$; hence $\mathrm{MSE}^\star > 0$ for every d. As $\lambda \to \infty$ the gain can cancel d only on the n_y directions that C measures, so b^2 has a positive limit when $(I - KC)\bar{d}$ cannot be made zero. (iii) Proposition 19 with Definition 18; in the linear case $z - P_{\mathrm{true}}x - \nu_{\mathrm{true}}$ obeys a stable linear recursion forced by $W_0 v_k + W_1 v_{k+1} - P_{\mathrm{true}} w_k$ only. □

Check (`m3_checks.py::check_model_mismatch`; damped oscillator, missing stiffness term $\Delta F = \epsilon\,[0\ 0; -1\ 0]$, $q = 0.05$, $\sigma_v = 0.03$, PRBS input). (i) the steady bias agrees with the formula to 10^{-8} (discrete, $\lambda = 0$ and 10^{-3}) and 10^{-6} (continuous Luenberger). (ii) at $\epsilon = 0.3$ (relative model error $|\Delta F x|/|Fx| = 0.29$), over $\lambda \in [0, 10^{-1}]$: $\|K\|$ rises $0.25 \to 1.19$, b^2 falls $5.24 \to 4.80 \cdot 10^{-2}$, $\operatorname{tr}\Sigma$ rises $2.0 \to 3.0 \cdot 10^{-3}$, MSE is U-shaped with $\mathrm{MSE}^\star = 5.09 \cdot 10^{-2}$; Monte Carlo agrees with $b^2 + \operatorname{tr}\Sigma$ within 1%. (iii) the true-plant KKL has $\mathrm{MSE} = 6.6 \cdot 10^{-3}$ and mean error 10^{-3} at every ϵ. Threshold: filters win at relative model error 0 and 0.05 ($\mathrm{MSE}^\star = 2.0, 4.2 \cdot 10^{-3}$), learned wins from 0.10 on ($10.0 \cdot 10^{-3}$ and above), one crossing.

What it predicts, and where it is tested (verdicts from `paper/generated/RESULTS.md`, the mechanical application of each node's pre-registered rule; model errors are the generated macros). With model error the C3 nodes measure, the proposition predicts a single crossing in the relative model error: filters win below it, learned observers above, per plant. On the exact-model side no condition reaches a learned-wins regime (C3-faults: none of 15 conditions; C3-noise-ct 0/6). On the incomplete-model side the learned observer beats the best *working* filter (a filter above climatology counts as failed) at every mismatch level, 10% (5/6 `id`, 4/6 `sequence_ood`), 20% (5/6, 5/6), 30% (5/6, 5/6), 50% (6/6, 5/6), under a missing term (C3-partial 5/6, 5/6) and in the realistic node (6/6, 6/6); the pre-registered crossover hypothesis "between 10% and 30%" therefore **fails** in the direction of the learned observer (first winning level 10%; the crossing lies below the smallest tested mismatch for the $\geq 4/6$ level rule, although duffing and battery are non-wins at 10%). Across the 36 model-wrong cells (`sequence_ood`, vs. the best working filter) the learned observer wins in 30/36. *Within* each system the outcome is ordered by model error: the six non-wins are duffing under every parameter mismatch (model error 0.02 at 30%), while duffing under the missing term (0.13) is a win; battery at 10% mismatch, while battery wins from 20% (model error 0.064); and quad_tank under the missing term (0.15), while quad_tank wins under parameter mismatch (0.69 at 30%); the systems with the largest errors (bioreactor 2.15, lorenz 0.71, vdp 0.66) are won at every level. *Across* systems the ordering does *not* hold: the smallest model error with a win (0.064, mismatch20/battery) is below the largest without one (0.152, partial/quad_tank). This is what the proposition allows (the threshold depends on the plant and the noise; the toy crosses at 0.05–0.10): the data are consistent with one crossing per plant, at plant-dependent thresholds that overlap in the band 0.06–0.15, not with a single cross-system threshold. This reading is post hoc, not a pre-registered test; TEST-C3 (frozen recipe, held-out plants) is **inconclusive** at the time of writing (6/18 jobs). The proposition is proven for linear plants; on the nonlinear benchmarks "learned wins iff the floor exceeds the fit error" is the reading of an **empirical** result. The "no model term" half of (iii) has one caveat: the phase-1 PDE term does use the (wrong) model; whether that helps or hurts is *not* established (C3-partial-modelfree: the partial model beats a model-free phase 1 with a CI excluding zero on 0/6 systems, so no claim that the partial model helps is made). Two further comparisons are needed before C3 can be read as a statement about KKL rather than about "data beats a wrong model": a Koopman/EDMDc Kalman observer and a grey-box (partial model

plus learned residual) EKF trained on the same data (`baselines_data.py`; not yet in the reported nodes).

3.2 Measurement and certification

Definition 24 (measured decomposition and predicted bound)**.** For a trained model, a stress condition and the steady window, in the σ-normalised norm of `metrics.py`, compute exactly: Δz_k by running the known filter on $v = \tilde{y} - \tilde{y}^{\text{nom}}$ (and $A_d^k \Delta z_0$ for the prior); Δc_k by running E on $\tilde{u}$ and u; and $\bar{J}_{z,k}, \bar{J}_{c,k}$ in (5) by Gauss quadrature along the traversed segments. Report the three terms of (5) (an exact attribution of the excess error) and the *realised effective gains* $g_z = \text{RMS}\|\bar{J}_z \Delta z\|/\text{RMS}\|\Delta z\|$, $g_c = \text{RMS}\|\bar{J}_c \Delta c\|/\text{RMS}\|\Delta c\|$. The *predicted bound* is $\widehat{\mathcal{B}} = \text{NRMSE}_{\text{nom}} + \hat{g}_z \text{RMS}\|\Delta z\| + \hat{g}_c \text{RMS}\|\Delta c\|$ with $\hat{g}_z, \hat{g}_c$ measured on *other* conditions (or the a priori $\kappa\|B\|\delta_y/\alpha$, $\sigma_v\|H\|_2$, $\ell_E \delta_u$ in place of the RMS terms).

With the gains of the same condition, $\widehat{\mathcal{B}} \geq \text{NRMSE}$ is the triangle inequality and tests nothing. With transferred gains it is a prediction, and the test is one-sided: a point above the diagonal falsifies the transferred gain (the decoder is steeper along this condition's paths); a point below the diagonal confirms nothing. *Status.* This procedure is evaluation-only on saved checkpoints (the filter, the encoder and the decoder are all available) and has not been run by any node in the current tree; until it is, the transferred-gain prediction is untested, and this section claims only (i), (ii), (iv) and the exact filter-level ratios above.

Remark 25 (what can be certified)**.** The result is a decomposition into measurable terms, not a certificate. *Exact* (no learning): $\kappa, \alpha, \gamma_y, \|H\|_2$, the latent deviation of any recorded fault, the path Jacobians on recorded data, and an upper bound on the SFM gain ℓ_E (product of layer norms, Lemma 15). *Lower bounds only*: any supremum of Jacobian norms estimated from samples. Exact Lipschitz constants are NP-hard to compute even for two-layer ReLU networks [17]; certified upper bounds (layer-norm products, LipSDP, 6) are valid but loose. On-data Jacobians are the wrong set for sensor faults: with $n_z > n_x$ the nominal latents lie near an n_x-dimensional manifold (times the context fibre) and $\hat{z} = \hat{z}^{\text{nom}} + \Delta z$ leaves it, where the decoder's slope is inductive bias; hence the traversed-segment Jacobians of Definition 24. The random-direction quotient l_{dec} of `train.phase1_diagnostics` estimates about $\|J\|_F/\sqrt{n_z}$ on the base decoder's data (check: $q_{0.95} = 0.81\|J\|_2$ for a linear decoder) and is not the constant of the bound. *Statistical*: the nominal error only through (4).

A Lemma D: the sampled-data observer

Lemma 26 (Lemma D)**.** *Let $\|e^{At}\| \leq \kappa e^{-\alpha t}$, $\rho_0 := e^{-\alpha dt}$, and let input and context be held on each $[t_k, t_{k+1})$. Let $w_0 := \int_0^{dt} \frac{\tau}{dt} e^{-\alpha\tau} d\tau = \frac{1-\rho_0(1+\alpha dt)}{\alpha^2 dt}$, $w_1 := \int_0^{dt} (1 - \frac{\tau}{dt}) e^{-\alpha\tau} d\tau = \frac{\alpha dt - 1 + \rho_0}{\alpha^2 dt}$, so $w_0 + w_1 = (1 - \rho_0)/\alpha$.*

(a) Exact step. *For y affine and g constant on the step, $\dot{z} = Az + By + g$ gives exactly $z_{k+1} = A_d z_k + W_0 y_k + W_1 y_{k+1} + W_g g$, with $A_d = e^{Adt}$, $W_0 = \int_0^{dt} (1 - \frac{s}{dt}) e^{A(dt-s)} B \, ds$, $W_1 = \int_0^{dt} \frac{s}{dt} e^{A(dt-s)} B \, ds$, $W_g = A^{-1}(A_d - I)$ (`design.latent_step`).*

(b) Interpolation. *With (A3) and the regularity clause, the clean iterate $\bar{z}_k$ from $\bar{z}_0 = z(0)$ satisfies $\sup_k \|\bar{z}_k - z(t_k)\| \leq e_{\text{disc}} := \frac{\kappa\|B\|M_2 dt^3}{12(1-\rho_0)} \leq \frac{\kappa\|B\|M_2 dt^2 e^{\alpha dt}}{12\alpha}$, also for switching inputs, since switches sit on the grid.*

(c) Merged kernels. $\|A_d^m W_0\| \le \kappa\|B\|\rho_0^m w_0$ *and* $\|A_d^m W_1\| \le \kappa\|B\|\rho_0^m w_1$. *Hence* $\sum_i \|G_i^{(k)}\| \le \kappa\|B\|/\alpha$ *for every* k *and* $\gamma_y \le \kappa\|B\|/\alpha$ *at every* dt, *and* $\gamma_y \to \int_0^\infty \|e^{As}B\|ds$ *as* $dt \to 0$. *More generally every sampled constant is bounded by its continuous counterpart:* $\rho_0^M = e^{-\alpha L}$, $c_d \le 1/\alpha$ *in (e),* $\gamma_y \le \kappa\|B\|/\alpha$.

(d) Injection: step-size condition. *For the deployed exponential midpoint step (an exponential integrator, 7, with an explicit Euler predictor)* $z_{k+1} = A_d z_k + W_0 y_k + W_1 y_{k+1} + W_g \Phi(z_k^h, c_k)$, $z_k^h = z_k + \frac{dt}{2}(Az_k + By_k + \Phi(z_k, c_k))$, *with* $\Phi(\cdot, c)$ ℓ_Φ*-Lipschitz, two solutions driven by the same* (y, c) *satisfy* $\|z_k - z_k'\| \le \kappa\rho^k\|z_0 - z_0'\|$,

$$\rho := \rho_0 + \kappa(1-\rho_0)\frac{\ell_\Phi \beta}{\alpha}, \qquad \beta := 1 + \tfrac{dt}{2}(\|A\| + \ell_\Phi).$$

So $\kappa\ell_\Phi\beta < \alpha$ *suffices; it tends to* $\kappa\ell_\Phi < \alpha$ *(Corollary 8) as* $dt \to 0$ *with* $-\ln\rho/dt \to \alpha - \kappa\ell_\Phi$, *and* $\beta \le 1.1$ *under* $\|A\|dt \le 0.1$ *(`design.choose_dt`). If the stationary latent* $\zeta_k = \hat{T}_0(x(t_k))$ *satisfies* $\zeta_{k+1} =$ (*clean step at* ζ_k) $+ \int_0^{dt} e^{A(dt-s)}R\,ds + q_k$, *with* $\|R\| \le \varepsilon_{\text{res}}$ *the PDE residual and* q_k *the interpolation/quadrature defect with* $\|A_d^m q_k\| \le \kappa\rho_0^m \bar{q}$, *and the step is fed samples with* $\|v_k\| \le \delta_y$, *then*

$$\|\hat{z}_k - \zeta_k\| \le \kappa\rho^k\|\hat{z}_0 - \zeta_0\| + \frac{\kappa\big(\|B\|(1+\ell_\Phi\frac{dt}{2})\delta_y + \varepsilon_{\text{res}}\big)}{\alpha - \kappa\ell_\Phi\beta} + \frac{\kappa\,\alpha\,\bar{q}}{(1-\rho_0)(\alpha - \kappa\ell_\Phi\beta)},$$

a single factor κ, *with* $\bar{q} = O(dt^3)$ *so the last term is* $O(dt^2)$.

(e) Deployed fading memory. *If* $u_j = u_j'$ *for* $j = k-M, \dots, k-1$, *the deployed map* $T_k^d(x;u) = \sum_{j\ge 0} A_d^j(W_0 y_{k-1-j} + W_1 y_{k-j})$ *satisfies* $\|T_k^d(x;u) - T_k^d(x;u')\| \le 2\kappa\|B\|\bar{y}\,\rho_0^M c_d$, $c_d := (w_0 + \rho_0 w_1)/(1-\rho_0) \le 1/\alpha$.

Proof. (a) Variation of constants over one step with the affine y inserted. (b) $\bar{z}_k - z(t_k) = -\int_0^{t_k} e^{A(t_k-s)}B(y-\ell)(s)ds$ with ℓ the piecewise-linear interpolant. On $[t_j, t_{j+1}]$ the input is constant, so $y \in C^2$ there, and the Peano form gives $\|y(t_j+\sigma) - \ell(t_j+\sigma)\| \le \frac{\sigma(dt-\sigma)}{2}M_2$ in any norm. Use $\|e^{A(t_k-s)}\| \le \kappa\rho_0^{k-1-j}$ on that interval and $\int_0^{dt}\frac{\sigma(dt-\sigma)}{2}d\sigma = \frac{dt^3}{12}$, sum the geometric series, and use $1-\rho_0 \ge \alpha dt\,\rho_0$. (c) $A_d^m W_0 = \int_0^{dt}(1-\frac{s}{dt})e^{A(mdt+dt-s)}B\,ds$, so $\|A_d^m W_0\| \le \kappa\|B\|\rho_0^m\int_0^{dt}(1-\frac{s}{dt})e^{-\alpha(dt-s)}ds = \kappa\|B\|\rho_0^m w_0$, and likewise for W_1. Then $\|H_0\| + \sum_{m\ge1}\|H_m\| \le \kappa\|B\|\big(w_1 + \sum_{m\ge1}(\rho_0^{m-1}w_0 + \rho_0^m w_1)\big) = \kappa\|B\|\frac{w_0+w_1}{1-\rho_0} = \frac{\kappa\|B\|}{\alpha}$, and the finite-$k$ coefficients are a sub-sum of the same terms. The limit holds because $H_m = e^{At_m}B\,dt + O(dt^2)$ with summable tails. (d) $\Delta_k = A_d^k\Delta_0 + \sum_{j<k}A_d^{k-1-j}W_g(\Phi(z_j^h) - \Phi(z_j'^h))$ with $\|z_j^h - z_j'^h\| \le \beta\|\Delta_j\|$ and the merged kernel $\|A_d^{k-1-j}W_g\| = \|\int_0^{dt}e^{A((k-j)dt-s)}ds\| \le \kappa\rho_0^{k-1-j}(1-\rho_0)/\alpha$. With $a_k = e^{\alpha t_k}\|\Delta_k\|$ and $g = \kappa(e^{\alpha dt}-1)\ell_\Phi\beta/\alpha$, $a_k \le \kappa a_0 + \sum_{j<k} g a_j$; the right side S_k obeys $S_{k+1} \le (1+g)S_k$, so $\|\Delta_k\| \le \kappa(\rho_0(1+g))^k\|\Delta_0\| = \kappa\rho^k\|\Delta_0\|$. If $\kappa\ell_\Phi < \alpha$ and $\|A\|dt \le 0.1$, then $\ell_\Phi dt < \alpha dt \le \|A\|dt$, so $\beta \le 1.1$. For the forced bound, unroll $e_k = \hat{z}_k - \zeta_k$; every forcing term is merged with the kernel that propagates it: $\|A_d^m(W_0 v_j + W_1 v_{j+1})\| \le \kappa\rho_0^m\|B\|(w_0+w_1)\delta_y$, $\|A_d^m\int_0^{dt}e^{A(dt-s)}R\,ds\| \le \kappa\rho_0^m(1-\rho_0)\varepsilon_{\text{res}}/\alpha$, the predictor mismatch $\le \beta\|e_j\| + \frac{dt}{2}\|B\|\delta_y$ enters through $A_d^m W_g$, and $\|A_d^m q_j\| \le \kappa\rho_0^m\bar{q}$. So $a_k \le \kappa a_0 + \sum_{j<k}\big(g a_j + \kappa e^{\alpha(j+1)dt}D\big)$ with $D = \frac{1-\rho_0}{\alpha}\big(\|B\|(1+\ell_\Phi\frac{dt}{2})\delta_y + \varepsilon_{\text{res}}\big) + \bar{q}$, hence $S_{k+1} \le (1+g)S_k + \kappa e^{\alpha(k+1)dt}D$ and $\|e_k\| \le \kappa\rho^k\|e_0\| + \kappa D/(1-\rho)$; insert $1-\rho = (1-\rho_0)(\alpha-\kappa\ell_\Phi\beta)/\alpha$. (e) The samples y_i coincide for $i \ge k-M$; bound by (c), summing from $j = M$ (W_0) and $j = M+1$ (W_1), with $\|y_i - y_i'\| \le 2\bar{y}$. □

Continuous-time interpretation. The latent step is exact for the held input, the context and any generated decoder weights are constant on each interval by construction, and the decoder runs at t_k only. The bounds are about the implemented observer, and the continuous formulas are their $dt \to 0$ limits; the sampled observer adds only e_{disc} and, with injection, the factor β.

Check (`m3_checks.py`). (b) `check_interpolation`: PRBS input, four step sizes, order 2.00, bound/error 3.6 (`check_theory.lemma_d` uses the same constant, bound/error 4.4–4.5). (c) `check_design_constants`: $\gamma_y = 1.074 \le \kappa\|B\|/\alpha = 1.491$ ($\int \|e^{As}B\| = 1.075$); $\gamma_y \le \kappa\|B\|/\alpha$ at $\alpha dt \in \{0.03, 0.3, 1, 3\}$ and for a non-normal A with $\kappa = 1.46$. The κ of a non-normal A is estimated by `spectral_bounds` on a 400-point grid, so those rows are consistency checks, not certificates; the deployed Butterworth A is normal and has $\kappa = 1$ exactly. (d) `check_midpoint`: for $\Phi = \ell z$ the per-step gain is below ρ (e.g. $0.99591 \le 0.99762$ at $\ell = 1.3$); 200 random linear injections: $\max \|\text{step}\|/\rho = 0.999$; for a non-normal A ($\kappa = 1.99$) $\max_k \|\text{step}^k\|/(\kappa\rho^k) = 0.92$, so a single κ suffices; $\Phi = -400z$, more stable in continuous time, gives discrete spectral radius 26.4 at $\ell dt = 8.2$, so the step-size part is needed; the numpy step equals `LatentStep` to 1.5×10^{-7}. (e) `m2_checks.py::lemma_d_constants`.

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